



GNSS Tracking through OneEdge

Application Note

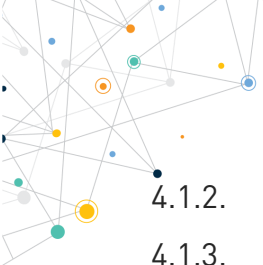
80654NT11966A Rev. 0 – 2021-11-24

APPLICABILITY TABLE

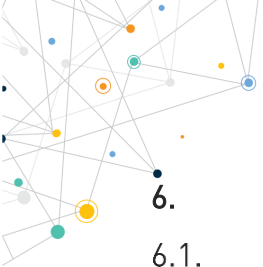
PRODUCTS	Platform Version ID
ME910C1-AU ME910C1-E1 ME910C1-E2 ME910C1-NA ME910C1-NV ME910C1-WW ML865C1-EA ML865C1-NA	30
ME910G1-W1 ME910G1-WW ME310G1-W1 ME310G1-WW ME310G1-W2 ML865G1-WW	37

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1. INTRODUCTION

1.1. Scope

This document describes the GNSS data management in OneEdge environment and how positioning and tracking applications can be developed on OneEdge.

The scope of the present document is to describe main details about location data management in OneEdge environment in order to publish positions on the map of the OneEdge IoT portal.

1.2. Audience

This document could be used also by Telit customers, especially system integrators, in order to develop applications based on Telit modules.

1.3. Contact Information, Support

For general contact, technical support services, technical questions and report of documentation errors contact Telit Technical Support at:

- TS-EMEA@telit.com
- TS-AMERICAS@telit.com
- TS-APAC@telit.com
- TS-SRD@telit.com
- TS-ONEEDGE@telit.com

Alternatively, use:

<https://www.telit.com/contact-us/>

For detailed information about where you can buy the Telit modules or for recommendations on accessories and components visit:

<https://www.telit.com>

Our aim is to make this guide as helpful as possible. Keep us informed of your comments and suggestions for improvements.

Telit appreciates the user feedback on our information.

1.4. Symbol Conventions



Danger: This information **MUST** be followed or catastrophic equipment failure or personal injury may occur.



Warning: Alerts the user on important steps about the module integration.



Note/Tip: Provides advice and suggestions that may be useful when integrating the module.



Electro-static Discharge: Notifies the user to take proper grounding precautions before handling the product.

Table 1: Symbol Conventions

All dates are in ISO 8601 format, that is YYYY-MM-DD.

1.5. Related Documents

- OneEdge Getting Started Guide: 1w0301585 Rev. 8
- Telit MEx10G1/ML865G1 GNSS Application Note: 80617NT11826A rev 7
- ME310G1/ME910G1/ML865G1 AT Commands Reference Guide: 80617ST10991A Rev. 9
- LWM2M AT Commands Reference Guide (ME310G1/ME910G1/ML865G1): 80617ST11022A Rev. 1
- LWM2M AT Commands Reference Guide (ME910C1/ML865C1): 80529ST10974A Rev. 4
- OMA Lightweight M2M Specification: OMA-TS-LightweightM2M-V1_0_2-20180209-A

2. OVERVIEW

This document describes how to manage all the information related to the GNSS position data on OneEdge environment.

Furthermore, also two typical use cases are explained: tracking and positioning applications.

It's highly recommended that the "OneEdge Getting started" guide is read and understood as precondition of this application note.

3. GNSS MANAGEMENT IN ONEEDGE ENVIRONMENT

3.1. Architecture

The image below shows the generic architecture of OneEdge environment.

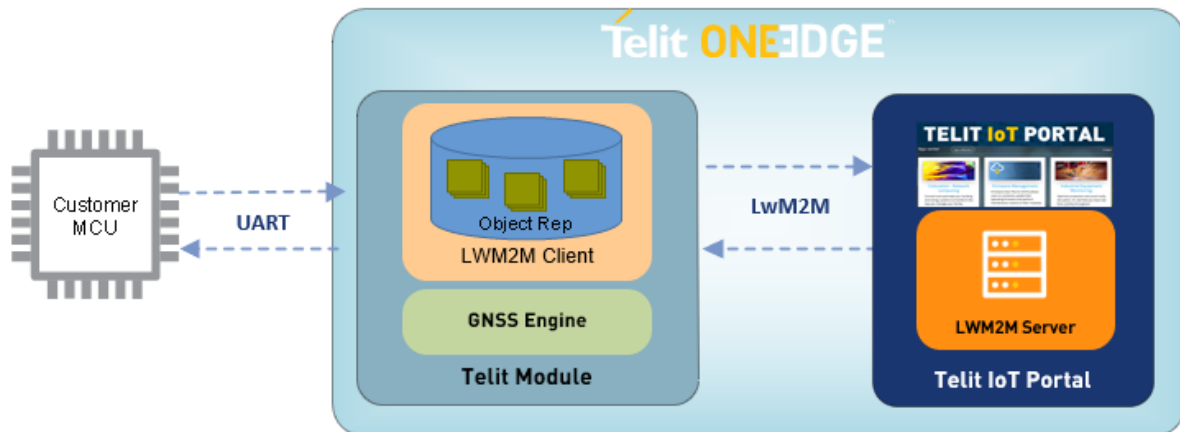


Figure 1: Main Architecture

The logical blocks for both the platforms on Applicability table, ME910C1 and MEx10G1 families, are represented in the Figure 2:

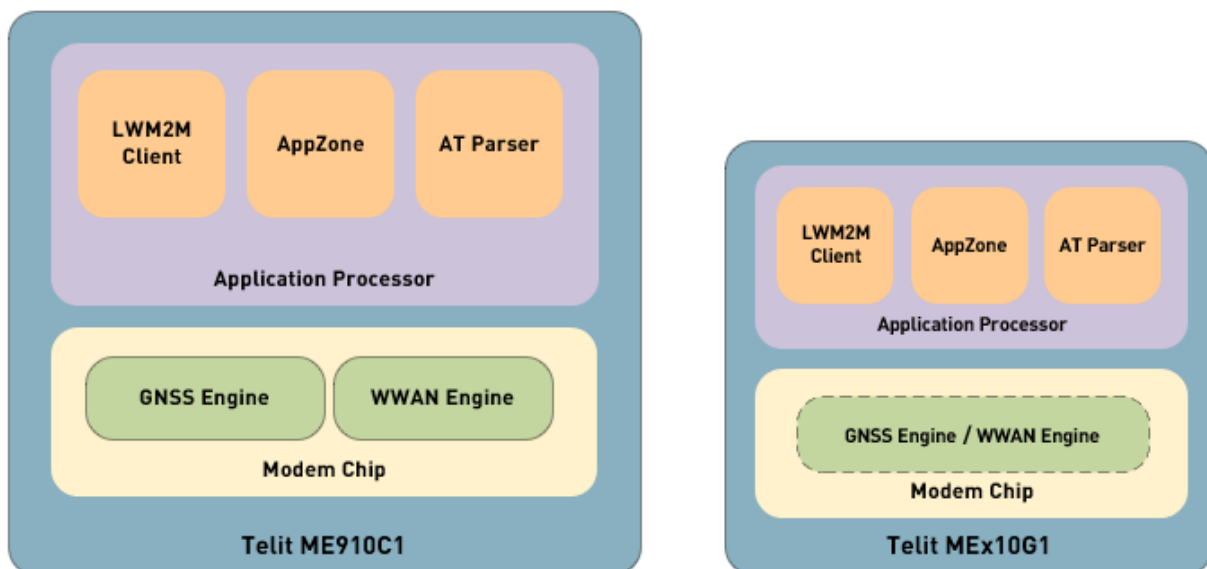


Figure 2: ME910C1 – MEx10G1 comparison

The details and differences are explained in next paragraphs.

3.1.1. Telit ME910C1 family

The modules of Telit ME910C1 family are capable of simultaneously perform the cellular (e.g.: cellular registration, paging, signaling and application data exchange) and the GNSS activities, as showed in the time flow diagram in Figure 3.

This is possible due to the fact that the hardware blocks on left side of Figure 2 (WWAN and GNSS) are separated and they can work together.

Related to the scope of this document, this capability allows a customer application to send the positioning data, collected through the GNSS antenna, to the OneEdge IoT portal in real time through the LwM2M protocol.

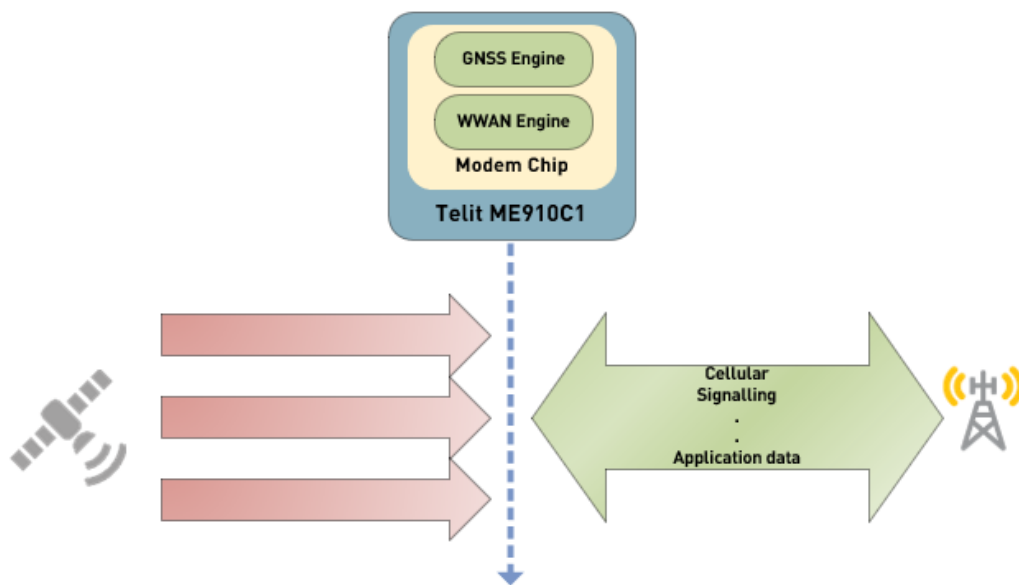


Figure 3: time flow diagram for ME910C1 family

3.1.2. Telit MEx10G1 family

For all the modules of MEx10G1 family, WWAN and GNSS blocks share the same hardware resources on modem chip (right side of Figure 2), therefore concurrent WWAN and GNSS operation are not possible.

It means that WWAN and GNSS operations are mutually exclusive: when the cellular part (WWAN) is running, the GNSS cannot be executed and vice versa (any WWAN activity cannot be performed as long as the GNSS session is active) as shown in Figure 4.

More details about the AT commands needed to manage executions of WWAN and GNSS parts are available on “Telit MEx10G1/ML865G1 GNSS Application Note: 80617NT11826A rev 7” guide.

This could be useful if the requirement of an application is to perform cellular data activity (e.g. LWM2M Client/Server data exchange) and GNSS activity (i.e.: gathering of the device position) on a cyclic basis.

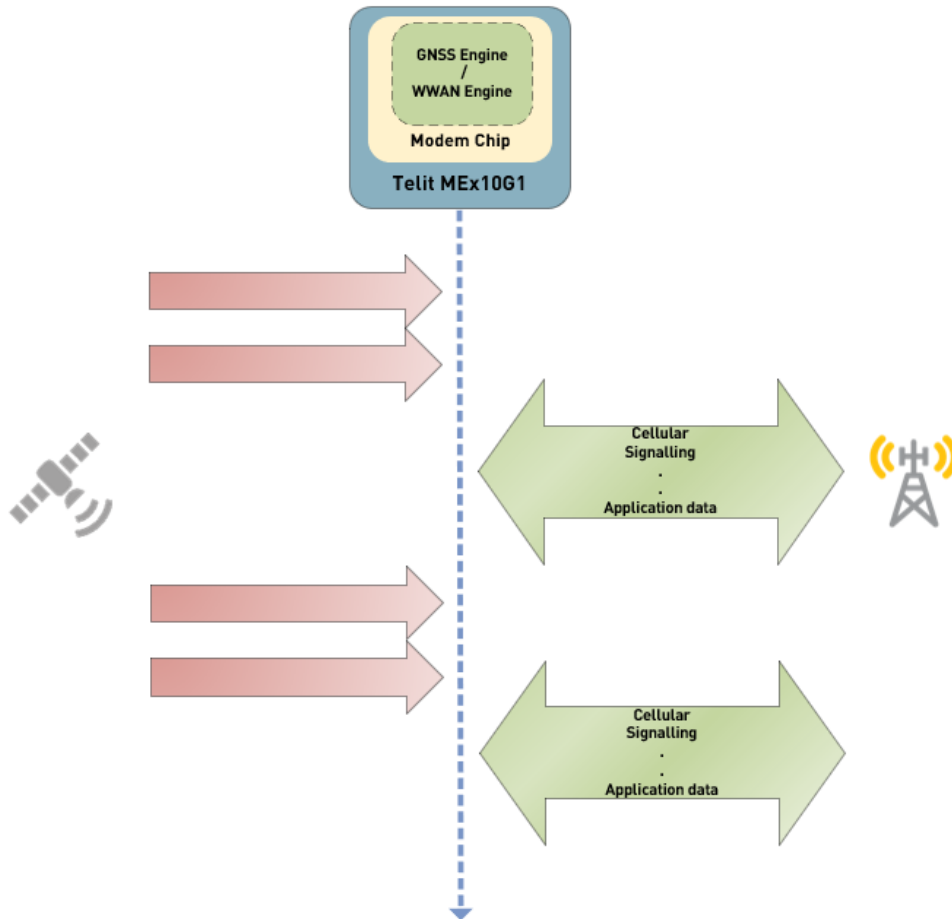


Figure 4: time flow diagram for MEx10G1 family

3.2. GNSS management in OneEdge

The GNSS receiver of Telit modules can be managed in OneEdge environment through a Telit proprietary LwM2M object described in next paragraph 3.2.1.

Warning: The LwM2M client doesn't manage position data if the GNSS receiver is activated through AT\$ commands (NMEA flow):



- AT\$GPSP=1 to activate it
- AT\$GPSP=0 to de-activate it

Be sure to use only LwM2M "Telit Services Management" object as described on paragraph 3.2.1

3.2.1. LwM2M “Telit Services Management” object

The proprietary “Telit Services Management” object (ID 33211) allows to set some useful parameters and features.

Telit Services Management		/33211			
Instance 0	/33211/0	Observe	Read	Write	Attributes
Service Enable	/33211/0/0	Observe	Read	Write	Attributes
Service Status	/33211/0/1	Observe	Read	Attributes	
External Observations Default Period	/33211/0/2	Observe	Read	Write	Attributes
Critical Observations	/33211/0/3	Observe	Read	Write	Attributes

Figure 5: Object 33211 – Portal View

Concerning the GNSS features, the involved resources are the following:

- **Service Enable (URI 33211/0/0):** the instance ID #0 of this multiple resource (res. ID 0) can be set equal to 1 or 0 in order to enable or disable the GNSS receiver. The AT commands on host application side are:
 - AT#LWM2MW=0,33211,0,0,0,1 to enable
 - AT#LWM2MW=0,33211,0,0,0,0 to disable

Of course, the GNSS receiver can be remotely enabled and disabled from the object browser (Figure 5) on Telit OneEdge IoT portal or through TR50 interface (lwm2m.device.oir.write API).

When a valid position fix is achieved (typically within **40-60** seconds), the updated GPS information become available in the resources of the OMA Location object (object ID 6, described in next paragraph [3.2.2](#)) at LwM2M client side.

The LwM2M client automatically fills the position information to the OMA Location object resources.

- **External Observation Default Period (URI 33211/0/2):** represents the internal polling cycle duration of the LwM2M client, with a **10** or **1800** seconds default value depending on the software version. It means that the LwM2M client gets the position information from the GNSS Engine every **XX** seconds. This parameter can be customized according to the application requirements.



Tip: The value of the “External observation default period” can be known reading the URI 33211/0/2 in the Telit OneEdge IoT Portal (Object browser section) or via AT Command directly from the LwM2M client.



Warning: “External observations default Period” is applied to all the internal resources monitored through the LwM2M client internal polling, such as RSSI and signal link quality (URI 4/0/2 and 4/0/3), not only to the GNSS resources.

3.2.2. Standard OMA Location object (object ID 6)

The LwM2M OMA “Location” object provides the ability to express a point (in 2D or 3D), a circle or a sphere.

This location information are used to indicate where the device is located at the indicated time. This point is mainly specified using latitude, longitude and timestamp. Furthermore, other optional information is available, like altitude, radius, velocity and speed.

The table below reports the list and description of the resources of OMA Location object:

Resource name	Resource Number	Mandatory	Type	Format	Description
Latitude	0	yes	Read only	Float	The decimal notation of latitude (e.g.: 39.22812462).
Longitude	1	yes	Read only	Float	The decimal notation of longitude (e.g.: 9.06913464).
Altitude	2	no	Read only	Float	The decimal notation of altitude in meters above sea level
Radius	3	no	Read only	Float	The size in meters of a circular area around a point geometry. It represents the accuracy of the position
Velocity	4	no	Read only	Opaque	The velocity of the LwM2M Client is defined in [3GPP-TS_23.032]
Timestamp	5	yes	Read Only	Time	The timestamp of when the location measurement was performed.

Speed	6	no	Read Only	Float	Speed is the time rate of change in position of a LwM2M Client without regard for direction: the scalar component of velocity.
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Table 1: Object 6 resources



Tip: The OMA Location object is always present in the LwM2M client and supported by default.

The host application can obtain the module position information reading the resources of this object, as explained in the section [3.2.4.3.1.Internal GNSS receiver](#).

The following limitations and constrains are linked to OMA Location object.

Whenever the GNSS receiver is enabled through the AT command `AT#LWM2MW=0,33211,0,0,0,1:`

- The LwM2M client periodically reports the location information from the GNSS engine on the resources of the OMA Location object;
- Since the resources are ReadOnly, according to the OMA LwM2M specification, the LwM2M server cannot modify the values of the resources;
- Even if the resources are ReadOnly, the host application on LwM2M client side cannot modify the values of the resources;



Warning: the OMA Location object is automatically filled by the LwM2M client only if the GNSS receiver is activated through the LwM2M command, not if it is managed by AT\$ commands.



Note: If the host application works with an external GNSS receiver or wants to post-process the position values, the IPSO Object "Location #3336" or a custom object can be used, please refer to the chapter [3.2.3](#).

The OMA Location object information are displayed on the OneEdge IoT portal through the object browser section. The Figure 6 represents an example:

Location /6					
Instance 0	/6/0	Observe	Read	Attributes	Delete
Latitude	/6/0/0	Observe	Read	Attributes	39.22822362
Longitude	/6/0/1	Observe	Read	Attributes	9.06922885
Altitude	/6/0/2	Observe	Read	Attributes	14.701259613037
Radius	/6/0/3	Observe	Read	Attributes	3.5355339050293
Velocity	/6/0/4	Observe	Read	Attributes	bearing=0, hSpeed=0
Timestamp	/6/0/5	Observe	Read	Attributes	2021-07-05T16:03:42Z
Speed	/6/0/6	Observe	Read	Attributes	0

Figure 6: OMA Object 6 "Location"

The paragraph [3.2.4](#) describes how to send Lwm2m data to the OneEdge IoT portal.

Reference:

- "OMA Lightweight M2M Specification: OMA-TS-LightweightM2M-V1_0_2-20180209-A" document: page 121.
- <https://github.com/OpenMobileAlliance/lwm2m-registry/blob/prod/6.xml>

3.2.2.1. MEx10G1 family limitations

According to paragraph 3.1.2, the host applications that use the modules of MEx10G1 family must perform a specific procedure to manage the succession between GNSS and WWAN sessions.

In particular, before to start the GNSS session, the related priority has to be set accordingly, as only activating the GNSS receiver is not sufficient to obtain the GNSS position on the OMA Location Object.

A basic host application step-by-step sequence to alternate GNSS and WWAN sessions is:

Precondition: set the GNSS Priority: AT\$GPSCFG=3,0

1. enable the GNSS receiver: AT#LWM2MW=0,33211,0,0,0,1
2. wait for GNSS fix reading OMA Location object values: this operation takes around 40/60 seconds, if it is the first loop
3. disable the GNSS receiver: AT#LWM2MW=0,33211,0,0,0,0
4. Now the host application can manage again WWAN activities, like sending data to Lwm2m server
5. restart from step 1 to get a new GNSS position

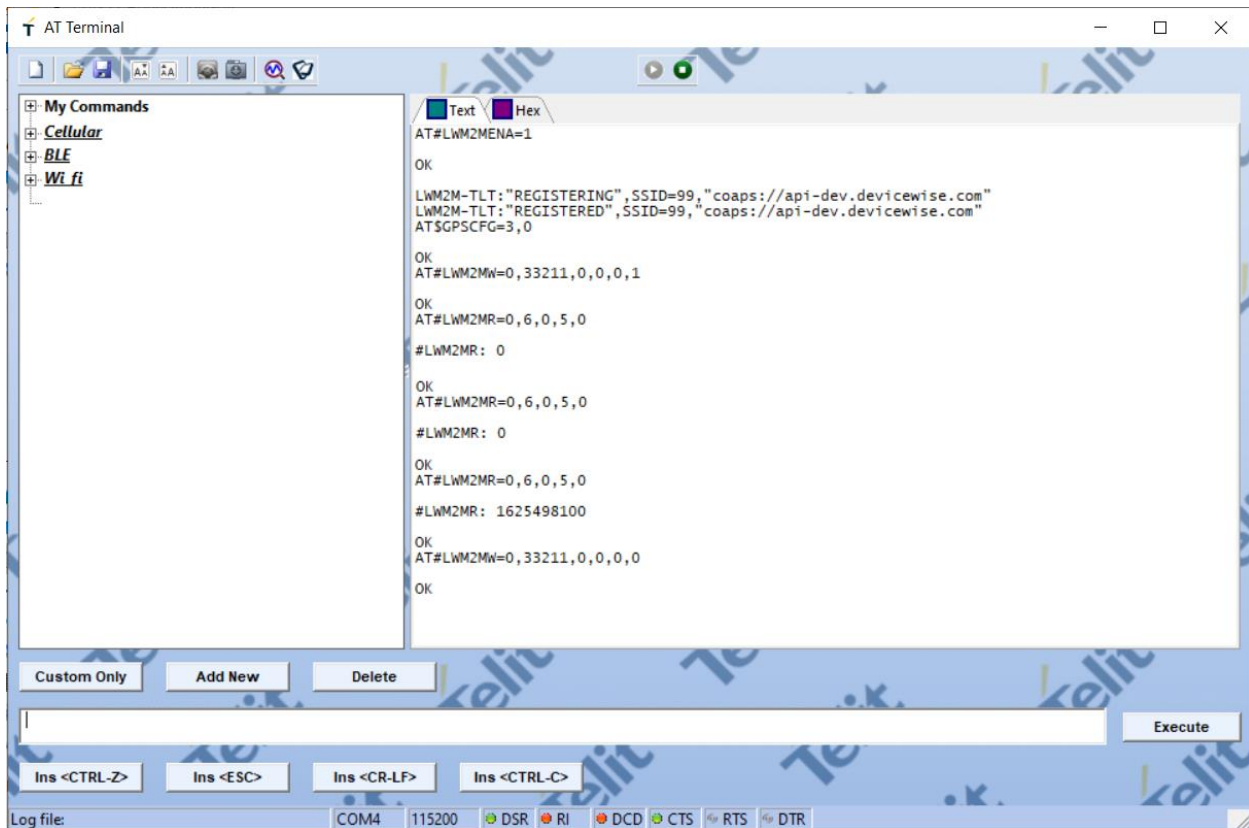


Figure 7: TATC Interface – command sequence



Tip: If the host application activates the LwM2M client at the beginning (sending AT#LWM2MENA=1), the connection between LwM2M client and server remains up and running in background.

3.2.3. OMA IPSO Location object (object ID 3336)

The IPSO Location object can contain generic GPS coordinates.

The table below reports the list and description of the resources of OMA IPSO Location object:

Resource name	Resource Number	Mandatory	Type	Format	Description
Numeric Latitude	6051	yes	Read only	Float	The decimal notation of latitude (e.g.: -43.5723).
Numeric Longitude	6052	yes	Read only	Float	The decimal notation of longitude (e.g.: 153.21760).
Numeric Uncertainty	6053	no	Read only	Float	The decimal notation of altitude in meters above sea level

Compass Direction	5705	no	Read only	Float	The measured compass direction
Velocity	5517	no	Read only	Opaque	The velocity of the LwM2M Client as defined in [3GPP-TS_23.032]
Timestamp	5518	yes	Read Only	Time	The timestamp of when the location measurement was performed.
Application Type	5750	no	Read/Write	String	The application type of the sensor or actuator as a string depending on the use case
Fractional Timestamp	6050	no	Read Only	Float	Fractional part of the timestamp when sub-second precision is used (e.g.: 0.23 for 230 ms).
Measurement Quality Indicator	6042	no	Read Only	Integer	Measurement quality indicator reported by a smart sensor: • 0: UNCHECKED No quality checks were done because they do not exist or cannot be applied; • 1: REJECTED WITH CERTAINTY The measured value is invalid; • 2: REJECTED WITH PROBABILITY The measured value is likely invalid; • 3: ACCEPTED BUT SUSPICIOUS The measured value is likely OK; • 4: ACCEPTED The measured value is OK; • 5-15: Reserved for future extensions; • 16-23: Vendor specific measurement quality.
Measurement Quality Level	6049	no	Read Only	Integer	Measurement quality level reported by a smart sensor. Quality level 100 means that the measurement has fully passed quality check algorithms. When the quality level is 0 it means that the measurement should certainly be rejected.

Table 2: IPSO Object 3336 resources

Even if the meaning of the resources of object 6 and 3336 are more or less the same, the main difference is that in OneEdge environment the object 6 is internally managed by the LwM2M client and the host cannot write the values on it.

On the other hand, the resources of object 3336 must be filled by the host application.

Some notes about this IPSO object are listed below:

- This object is suitable to contain positioning data coming from an external GNSS receiver: the host application should get the values from the external source and write them to the resources of the object 3336;
- This object is suitable to contain pre-processed positioning data coming from the internal GNSS receiver: the host application should get the values from object 6, optionally it could elaborate and apply some corrections, and then write them to the resources of object 3336.

Tip: Some examples of pre-processing of data on host application could be:



- send position only if latitude and longitude change in order to avoid publishing multiple positions in case the device doesn't move;
- send position only if latitude and longitude change by a defined distance;
- send position only if timestamp changes;
- send position only if accuracy is under a threshold;
- add a correction factor to altitude
- etc.



Warning: the object 3336 is not present by default in the LwM2M client, so it must be loaded on the module before to start with the host application developments. The xml file can be downloaded from OMA web site.

Reference:

<https://github.com/OpenMobileAlliance/lwm2m-registry/blob/prod/3336.xml>

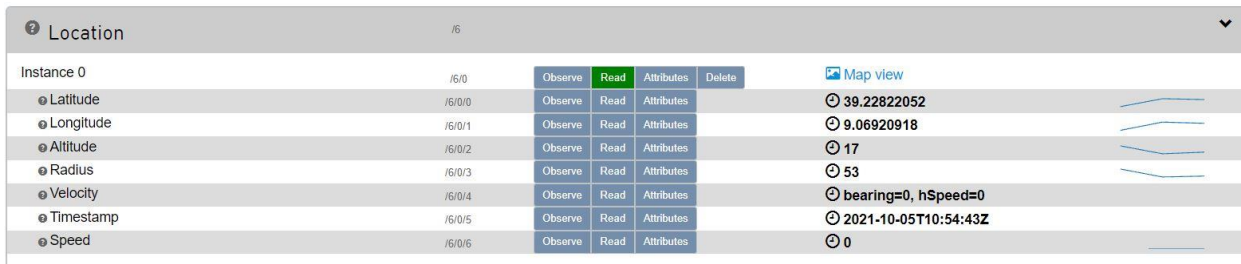
3.2.4. Sending a position to the OneEdge IoT portal

Once the GNSS receiver has been activated through the LwM2M command "AT#LWM2MW=0,33211,0,0,0,1", the LwM2M client starts to write the positioning data on the resources of the object 6. The information is updated every 10 or 1800 seconds (according to the "External observations default Period").

Then the positioning data can be sent to the LwM2M server in several ways, as described below.

3.2.4.1. Manual pulling

The user can perform a READ operation by pressing the “Read” button on the object browser section every time he wants to see the data updated in the IoT Portal



Location		/6				
Instance 0	/6/0	Observe	Read	Attributes	Delete	Map view
Latitude	/6/0/0	Observe	Read	Attributes		39.22822052
Longitude	/6/0/1	Observe	Read	Attributes		9.06920918
Altitude	/6/0/2	Observe	Read	Attributes		17
Radius	/6/0/3	Observe	Read	Attributes		53
Velocity	/6/0/4	Observe	Read	Attributes		bearing=0, hSpeed=0
Timestamp	/6/0/5	Observe	Read	Attributes		2021-10-05T10:54:43Z
Speed	/6/0/6	Observe	Read	Attributes		0

Figure 8: Manual read on object 6

The same action can be performed by an external application through the TR50 `lwm2m.device.oir.read` api.

For example:

Request

```
{
  "1": {
    "command": "lwm2m.device.oir.read",
    "params": {
      "thingKey": "353081094060194",
      "objId": 6,
      "instId": 0
    }
  }
}
```

Response

```
{
  "1": {
    "success": true,
    "params": {
      "result": {
```

```
"6": {
  "0": {
    "0": {
      "ts": "2021-10-19T14:21:31.17185262Z",
      "value": 39.22823491
    },
    "1": {
      "ts": "2021-10-19T14:21:31.17185262Z",
      "value": 9.0692463
    },
    "2": {
      "ts": "2021-10-19T14:21:31.17185262Z",
      "value": 18
    },
    "3": {
      "ts": "2021-10-19T14:21:31.17185262Z",
      "value": 3
    },
    "4": {
      "ts": "2021-10-19T14:21:31.17185262Z",
      "value": {
        "bearing": 0,
        "hSpeed": 0
      }
    },
    "5": {
      "ts": "2021-10-19T14:21:31.17185262Z",
      "value": "2021-10-19T14:18:54Z"
    },
    "6": {
      "ts": "2021-10-19T14:21:31.17185262Z",
      "value": 0
    }
  }
}
```

```
}  
}  
}  
}
```

Reference:

https://docs.devicewise.com/Content/Products/loT_Portal_API_Reference_Guide/APIs/lwm2m/lwm2m.device.oir.read.htm

3.2.4.2. Automatic pulling

In OneEdge environment, the data stored in the object 6 can be read by the IoT portal also configuring the Device profile, as described below:

- **Read on register:** if this option is set equal to “YES”, every time the Lwm2M Client completes the registration to the Lwm2M Server, the OneEdge IoT Portal performs an automatic read, so the information stored in the resources of object 6 on Client side are sent to the OneEdge IoT portal;
- **Read on update:** if this option is set equal to “YES”, every time the Lwm2M Client completes the registration update to the Lwm2M Server, the Portal performs an automatic read, so the information stored in the resources of object 6 on Client side are sent to the OneEdge IoT portal. This option is strictly related to the lifetime parameter: the frequency of the registration update depends on its value;
- **Observe:** if this option is set to “YES”, every time the positioning data change in the object 6, the Lwm2M Client sends a notification with the updated values to the Lwm2M Server and the OneEdge IoT portal shows the new positioning value.

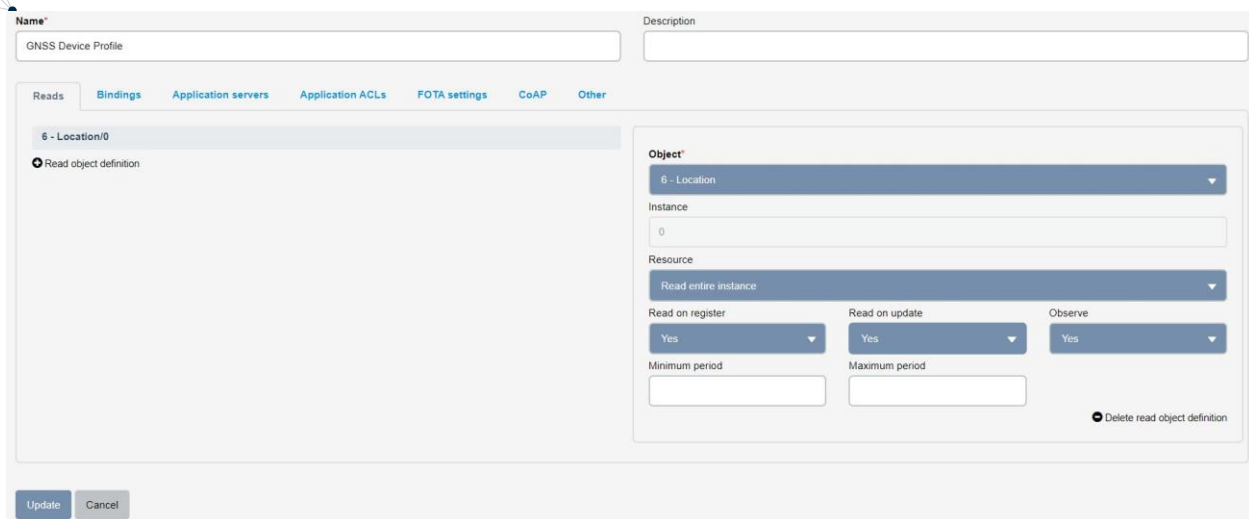


Figure 9: Device profile settings – object 6

3.2.4.3. Object ID 3336

All the methods described above can also be applied to obtain the positioning data in the portal section using the object ID 3336, but some preconditions are needed.

3.2.4.3.1. Internal GNSS receiver

When the GNSS receiver is active, a valid position is reported to object 6 and updated constantly.

The host application must transfer these data from the object 6 to the object 3336, making a sort of “copy&paste” operation, as described in the following step flow:

- a) Read the values contained in the resources of the object 6 with the AT command

```
AT#LWM2MOBJGET=0,6,0
{"bn":"/6/0/",
 "e":[
  {"n":"0","v":39.228206},
  {"n":"1","v":9.069231},
  {"n":"2","v":29.000000},
  {"n":"3","v":20.000000},
  {"n":"4","sv":"AAAAAA=="},
  {"n":"5","v":1633428348},
  {"n":"6","v":0.000000}]
}
```

- b) Perform any required elaboration and check: i.e. perform step c only if latitude and longitude change
- c) write the values from the object 6 (only the resources useful for the customer application purpose) into the object 3336 through the command

```
AT#LWM2MOBJSET=0,3336,0
{"bn":"/3336/0/",
```

```
"e":[
{"n":"6051","v":39.228206},
{"n":"6052","v":9.069231},
{"n":"6053","v":20.0},
{"n":"5518","v":1633428348},
{"n":"5517","sv":"AAAAAA=="}]
}<Ctrl-Z>
```

Warning: the AT commands **AT#LWM2MOBJGET** and **AT#LWM2MOBJSET** are available only from 30.01.xx0 - M0B.xx0005 and 37.00.xx4 - M0C.xx0003 SW versions. In older SW versions the same actions can be performed through single operations on each resource with **AT#LWM2MR** and **AT#LWM2MSET** commands:

```
AT#LWM2MR=0,6,0,0,0
```

```
AT#LWM2MR=0,6,0,1,0
```

```
AT#LWM2MR=0,6,0,2,0
```

```
AT#LWM2MR=0,6,0,3,0
```

```
AT#LWM2MR=0,6,0,4,0
```

```
AT#LWM2MR=0,6,0,5,0
```

```
AT#LWM2MR=0,6,0,6,0
```



```
AT#LWM2MSET=1,3336,0,6051,0,<value> (float)
```

```
AT#LWM2MSET=1,3336,0,6052,0,<value> (float)
```

```
AT#LWM2MSET=1,3336,0,6053,0,<value> (float)
```

```
AT#LWM2MSET=5,3336,0,5518,0,<value> (time)
```

```
AT#LWM2MSET=3,3336,0,5517,0,<value> (opaque)
```

Now the values are available in the resources of the object 3336 at client side and they can be sent to the Lwm2M portal using the methods described in chapters [3.2.4.1](#) and [3.2.4.2](#).

3.2.4.3.2. External GNSS receiver

The object 3336 is especially dedicated for the customer applications equipped with an external GNSS receiver.

In these cases, the positioning information obtained from the external GNSS receiver must be processed by the host application and converted in a format (string, float, integer and so on) compatible with the resources definition.

Once the data are ready and they are written on the object 3336, the client can manage the data sending in accordance with the setting defined by the user (chapters 3.2.4.1 and 3.2.4.2.).

3.2.5. Position management on OneEdge IoT portal

The OneEdge IoT Portal has a specific section named “Location history” that contains the positioning data showed in two different ways:

- **Map view (at thing level):** The Map tab of the Location history shows the path as a line on a map with the points traveled by the thing. The last known position is marked on the map through a dialog info box and its information is reported in the table under the map (Figure 10).
- **Table view (at thing level):** it contains the same information in a table format, there are as many entries in the table as there are points in the map.

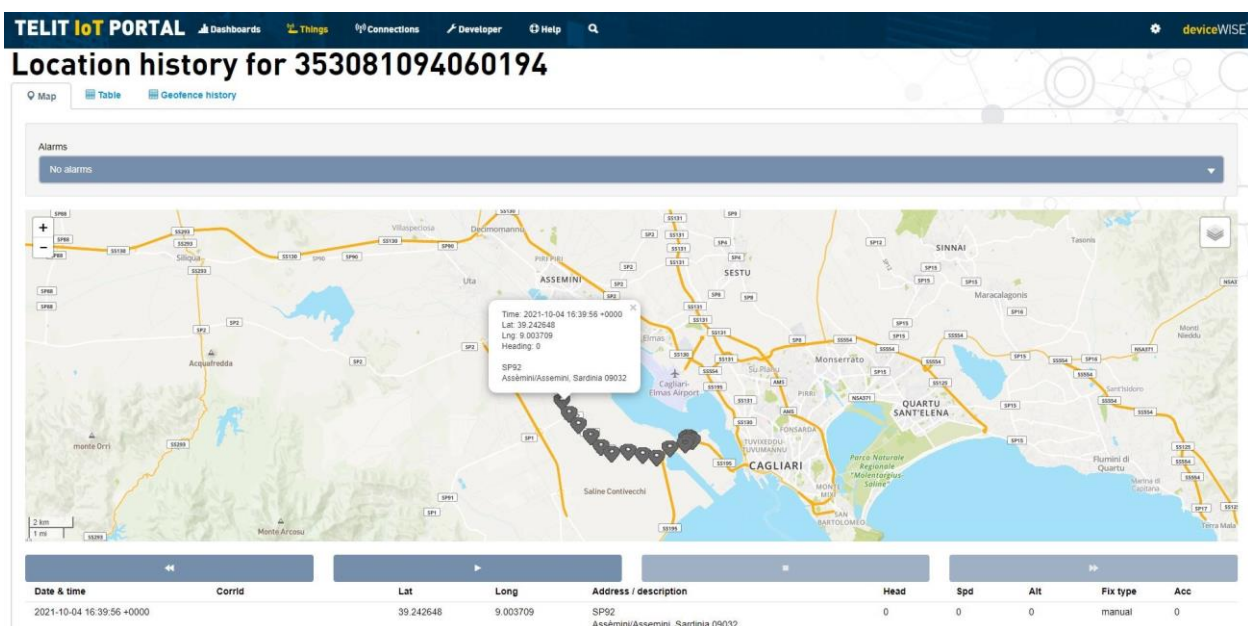


Figure 10: Location history – Map View

TELIT IoT PORTAL Dashboards Things Connections Developer Help

Things > 353081094060194 > Location history for 353081094060194

Location history for 353081094060194

Map Table Geofence history Download CSV

Date & time	Corrid	Latitude	Longitude	Delta	Address / description	Heading	Speed	Altitude	Fix type	Accuracy	State	Country
2021-10-04 16:34:02 +0000		39.226338	9.068646	121 meters	Statale 195 Sulcitana Cagliari Sardinia IT	0	0	0 meters	manual	0 meters	Sardinia	IT
2021-10-04 16:34:32 +0000		39.226756	9.071283	232 meters	Statale 195 Sulcitana Cagliari Sardinia IT	0	0	0 meters	manual	0 meters	Sardinia	IT
2021-10-04 16:34:59 +0000		39.225437	9.067643	346 meters	Statale 195 Sulcitana Cagliari Sardinia IT	0	0	0 meters	manual	0 meters	Sardinia	IT
2021-10-04 16:35:29 +0000		39.223637	9.059621	720 meters	Statale 195 Sulcitana Cagliari Sardinia IT	0	0	0 meters	manual	0 meters	Sardinia	IT
2021-10-04 16:35:59 +0000		39.219893	9.052827	718 meters	unnamed road Cagliari Sardinia IT	0	0	0 meters	manual	0 meters	Sardinia	IT
2021-10-04 16:36:29 +0000		39.221181	9.045457	651 meters	Saline Contevichi Cagliari Sardinia IT	0	0	0 meters	manual	0 meters	Sardinia	IT
2021-10-04 16:36:59 +0000		39.222334	9.038053	651 meters	Saline Contevichi Cagliari Sardinia IT	0	0	0 meters	manual	0 meters	Sardinia	IT
2021-10-04 16:37:29 +0000		39.221554	9.029512	741 meters	Saline Contevichi Cagliari Sardinia IT	0	0	0 meters	manual	0 meters	Sardinia	IT
2021-10-04 16:37:59 +0000		39.224311	9.02406	561 meters	Saline Contevichi Assemini/Assemini Sardinia 09032 IT	0	0	0 meters	manual	0 meters	Sardinia	IT
2021-10-04 16:38:26 +0000		39.228558	9.018887	649 meters	Saline Contevichi Assemini/Assemini Sardinia 09032 IT	0	0	0 meters	manual	0 meters	Sardinia	IT
2021-10-04 16:38:56 +0000		39.23293	9.012213	753 meters	SP92 Assemini/Assemini Sardinia 09032 IT	0	0	0 meters	manual	0 meters	Sardinia	IT
2021-10-04 16:39:26 +0000		39.237471	9.007517	647 meters	SP92 Assemini/Assemini Sardinia 09032 IT	0	0	0 meters	manual	0 meters	Sardinia	IT
2021-10-04 16:39:56 +0000		39.242648	9.003709	663 meters	SP92 Assemini/Assemini Sardinia 09032 IT	0	0	0 meters	manual	0 meters	Sardinia	IT

Figure 11: Location history – Table View

The location information are also available in the main page:

- **Fleet view (at org level):** it displays the locations of the whole cellular devices in a global map. Using the zoom button is possible to focus on a specific area with details of each device, as shown by the Figure 12. The OneEdge IoT portal is properly configured to publish the thing positions in the map according to the data received by object 6.

The device profile settings, in particular related to the object 6 and lifetime parameter, together with the External Observation Default Period (URI 33211/0/2) affect the location history data.

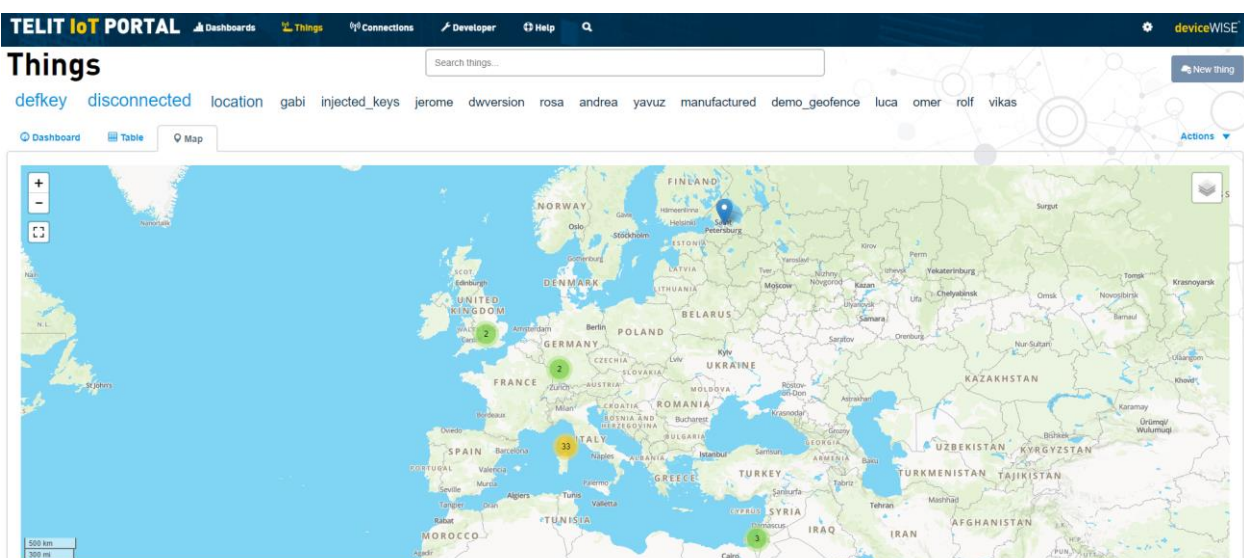


Figure 12: Global map view

The Location History section reports the positioning data coming from different fix types:

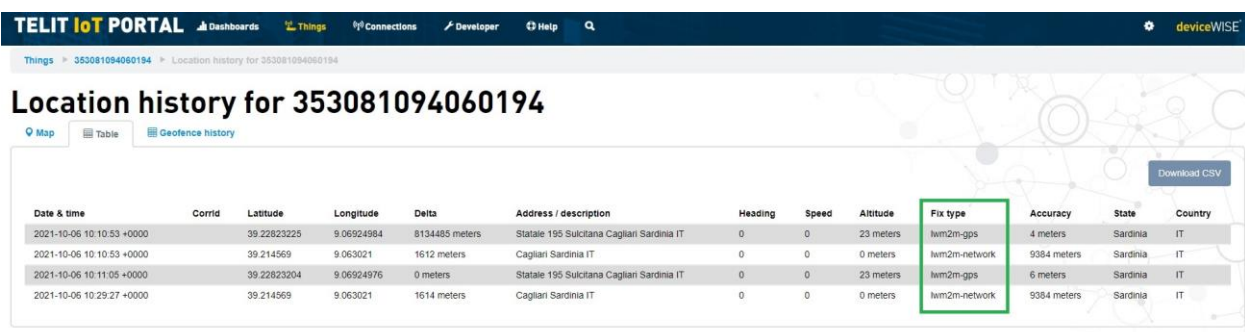
- **Manual:** a manually defined location provided by the user (through location settings, API functions or triggers);
- **network:** location acquired from the SIM card (Connections information);
- **lwm2m-gps:** location acquired from a GPS satellite system (LwM2M object 6);
- **lwm2m-network:** location acquired from a network location service (LwM2M object 4).

All the fix types can be active at the same time, but some considerations must be taken.

For example, if the OneEdge IoT Portal receives data for both lwm2m-gps and lwm2m-network positioning fix types, both the locations are displayed on the map, but with the appropriate accuracy.

Usually lwm2m-gps locations have higher accuracy than lwm2m-network ones, as showed in Figure 13.

This is because the location information coming from the network depends on the serving cellular cell and power resolution through a third party service, which in most of cases is less accurate than a GNSS receiver.



Date & time	Corrid	Latitude	Longitude	Delta	Address / description	Heading	Speed	Altitude	Fix type	Accuracy	State	Country
2021-10-06 10:10:53 +0000	39.22823225	9.06924984	8134485 meters		Statale 195 Sulcitana Cagliari Sardinia IT	0	0	23 meters	lwm2m-gps	4 meters	Sardinia	IT
2021-10-06 10:10:53 +0000	39.214569	9.063021	1612 meters		Cagliari Sardinia IT	0	0	0 meters	lwm2m-network	9384 meters	Sardinia	IT
2021-10-06 10:11:05 +0000	39.22823204	9.06924976	0 meters		Statale 195 Sulcitana Cagliari Sardinia IT	0	0	23 meters	lwm2m-gps	6 meters	Sardinia	IT
2021-10-06 10:29:27 +0000	39.214569	9.063021	1614 meters		Cagliari Sardinia IT	0	0	0 meters	lwm2m-network	9384 meters	Sardinia	IT

Figure 13: Fix types

In order to avoid this mixed information, the User can disable the unneeded fix types from the Locations tab of the Thing definitions settings (Developer section of IoT Portal).

If, for some reason, the customer wants to show in the Local history the information coming from the object 3336 instead of object 6, a specific trigger must be implemented.

The location.publish trigger action is used to publish a thing's location with latitude/longitude. Speed, direction, and altitude are optional parameters.

A simple trigger like the following one can satisfy the customer requirements:

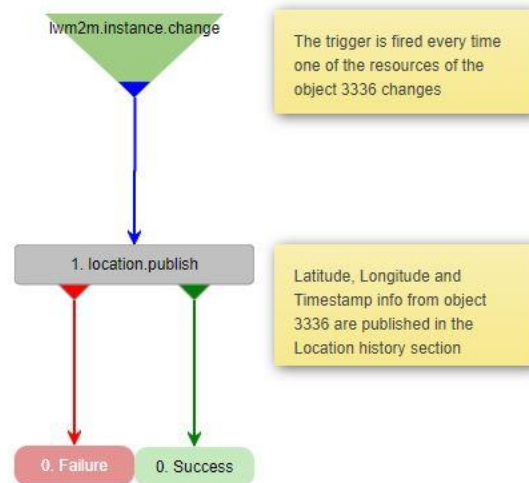


Figure 14: Trigger

The positions published by the location.publish trigger connector are reported in the Location history with “Manual” fix type.

Debounce is another optional parameter of the location.publish trigger connector, which proves to be very useful to tune the values being published on the Location history.

The **debounce** value, defined as an integer, is the value (in meters) used to determine if the location publish should be suppressed, from the table and also graphical view, to avoid drifting in a small area due to inaccuracies.

The default value is 50 meters when it is not specified, indicating that if the location published is less than 50 meters from the previous location, then it will be ignored. Setting it to 0 disables debouncing.



Tip: Setting the debounce value in a proper way helps to reduce API count.

References:

https://docs.devicewise.com/Content/Products/Management_Portal/Developer/Triggers/Actions/location.publish-action.htm?Highlight=location.publish

3.2.6. How the position is obtained

When the GNSS receiver is active, it receives the signal broadcasted by the satellites on the sky.

Sometimes, in case of bad conditions like poor antenna quality, adverse weather conditions, building canyoning or tunnels environment, the number of detected satellites and their signal power level could be not good enough to allow a valid position calculation.

The validity of a GNSS position is called as “position fix” or simply “fix”.



Tip: referring to classic GNSS management, in \$GPRMC NMEA sentence, the fix validity is represented through the status field that can be equal to “A” in case of valid position and “V” for fix not achieved.

Since the fix validity information is not present in any of the resources of OMA Location object, an host application has to implement a different strategy to evaluate it.

3.2.7. GNSS fix evaluation

As the default values of the OMA Location object after every modem power up are all equal to “0”, this means that the first position is a dummy one, even if a fix was present before the power cycle.

Also when the GNSS receiver is enabled but the location fix has not been achieved yet, the resources are equal to 0 (also the timestamp).

As soon as the fix is obtained, the LwM2M client fills the valid position information on the OMA Location object and a host application can detect the variation of the timestamp from “0” to current time.

The AT command sent to get the position information receives a response equal to 0, as showed in Figure 15.

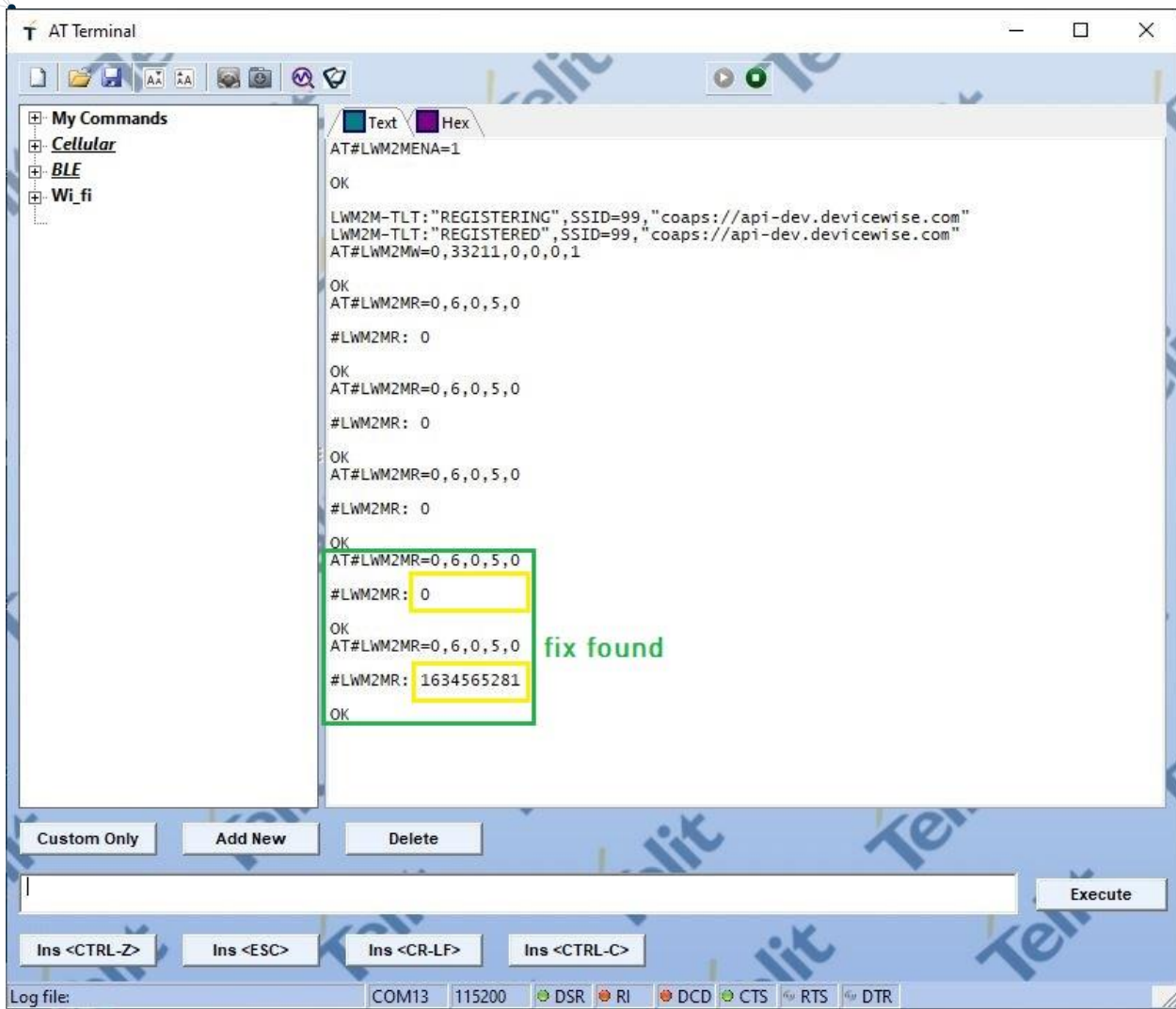


Figure 15: ME910C1 Family – waiting for fix

Checking the timestamp resource (6/0/5), the host application can know if the fix is available or not. If the position is valid (timestamp different from 0) the host application can read all the resources of the object 6 to take the position information.

Sometimes it is possible that the GPS signal can be lost, for example inside a tunnel, in an underground parking or in the cargo hold. In this case the host application can evaluate if the position information is valid or not by comparing current time and last one.

If all the resources of the object 6 have the same value of the previous ones and this condition appears for a certain time window, it means that the position is not more valid and the signal has been lost.

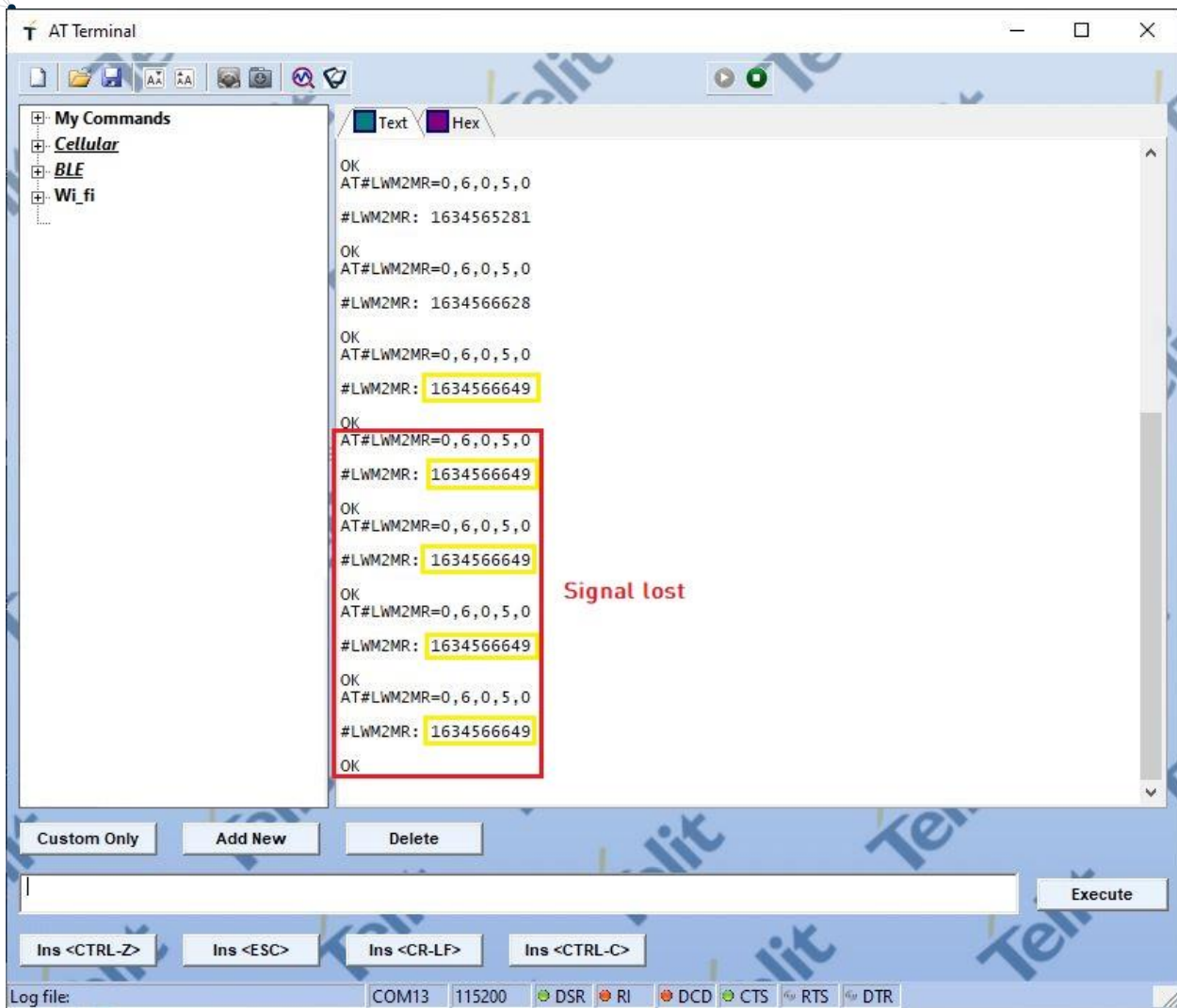


Figure 16: Signal lost example

Basically, if the GNSS receiver loses the fix because the device goes from an open sky environment into a tunnel, it stops updating the position values on the OMA Location object, so the timestamp value on resource 6/0/5 does not grow compared to the current time of the device.

Tip: on Telit devices, the current time, in Epoch Unix format, can be returned by following command:

AT#LWM2MR=0,3,0,13,0
 #LWM2MR: 1638354969
 OK

Another interesting piece of information related to the quality and validity of the location information is the **Radius/Accuracy** (resource 6/0/3): values above 50-100 meters could indicate poor signal conditions.

3.2.7.1. AT#LWM2MOBJGET command

If the module supports the AT#LWM2MOBJGET command, the process to evaluate the fix validity is easier:

- If the values of all the resources are equal to 0, the module is waiting for the first fix

```
AT#LWM2MOBJGET=0,6,0
{"bn":"/6/0/",
 "e":[
  {"n":"0","v":0.000000},
  {"n":"1","v":0.000000},
  {"n":"2","v":0.000000},
  {"n":"3","v":0.000000},
  {"n":"4","sv":"AAAAAA=="},
  {"n":"5","v":0},
  {"n":"6","v":0.000000}]
```

- If all the values are equal to the previous ones, the GPS signal has been lost.

4. TRACKING USE CASE

Applications based on GNSS can cover many scopes, in particular related to moving objects monitoring.

Some use cases could be:

- Maritime navigation: vessels monitoring and fleet management;
- Port Automation: shipping hubs can improve their operating efficiency by tracking the movement and placement of containers about their yards;
- Parking Automation: the monitoring of the parked vehicles;
- Vehicle rental, bus line and taxis: collected information provides indications about locations, stops and distance covered by each item; they can be monitored and processed for further analysis.
- Precision agriculture: GNSS-based applications are used to support farm planning, field mapping, soil sampling, tractor route monitoring, etc;

The vehicle's locations are collected on LwM2M client side and transmitted periodically to the LwM2M server over a GSM cellular network. The customer accesses to the information by logging on to the OneEdge IoT Portal, which requires a monthly fee, or by receiving the data directly on a smartphone. The positions are updated every few minutes, according with the system specification.

In the following paragraphs a specific tracking use case is described.

4.1. High-level requirements of the application

The tracking device sends geo-spatial information to the OneEdge IoT portal at high frequency, for example every 2 minutes.

Below the main requirements/features:

4.1.1. Power Supply

- **[REQ #1] The device is powered for all the tracking time**

Since the tracking application runs for all the time of the tracking, the device must be powered by a stable source of electricity, like batteries on vehicles.

Furthermore, a tracking device usually doesn't need to implement aggressive power save modes.

4.1.2. Device to Portal direction

- **[REQ #2] The tracking device sends geo-spatial information from internal GNSS receiver to the OneEdge IoT portal at specific frequency**

The host application manages a default sending rate of 2 minutes.

Geo-spatial information are the following:

- ✓ Latitude
- ✓ Longitude
- ✓ Altitude
- ✓ Radius/Accuracy
- ✓ Velocity/Heading
- ✓ Timestamp
- ✓ Speed

4.1.3. Portal view

- **[REQ #3] Map**

The geo-spatial information are plotted on a map on the OneEdge IoT portal.

- **[REQ #4] Table**

The geo-spatial information are available in text format list on the OneEdge IoT Portal or downloadable in a CSV file.

4.2. Data flow diagram

The image below shows the flow of AT commands, LwM2M messages and events across the actors involved in the tracking use case.

The vertical arrows indicate the time, starting from the top to the bottom.

All the three features are represented:

- the registration phase which is a one-time action;
- getting position from GNSS engine (for both families);
- sending position to the portal to show the geo-spatial information.

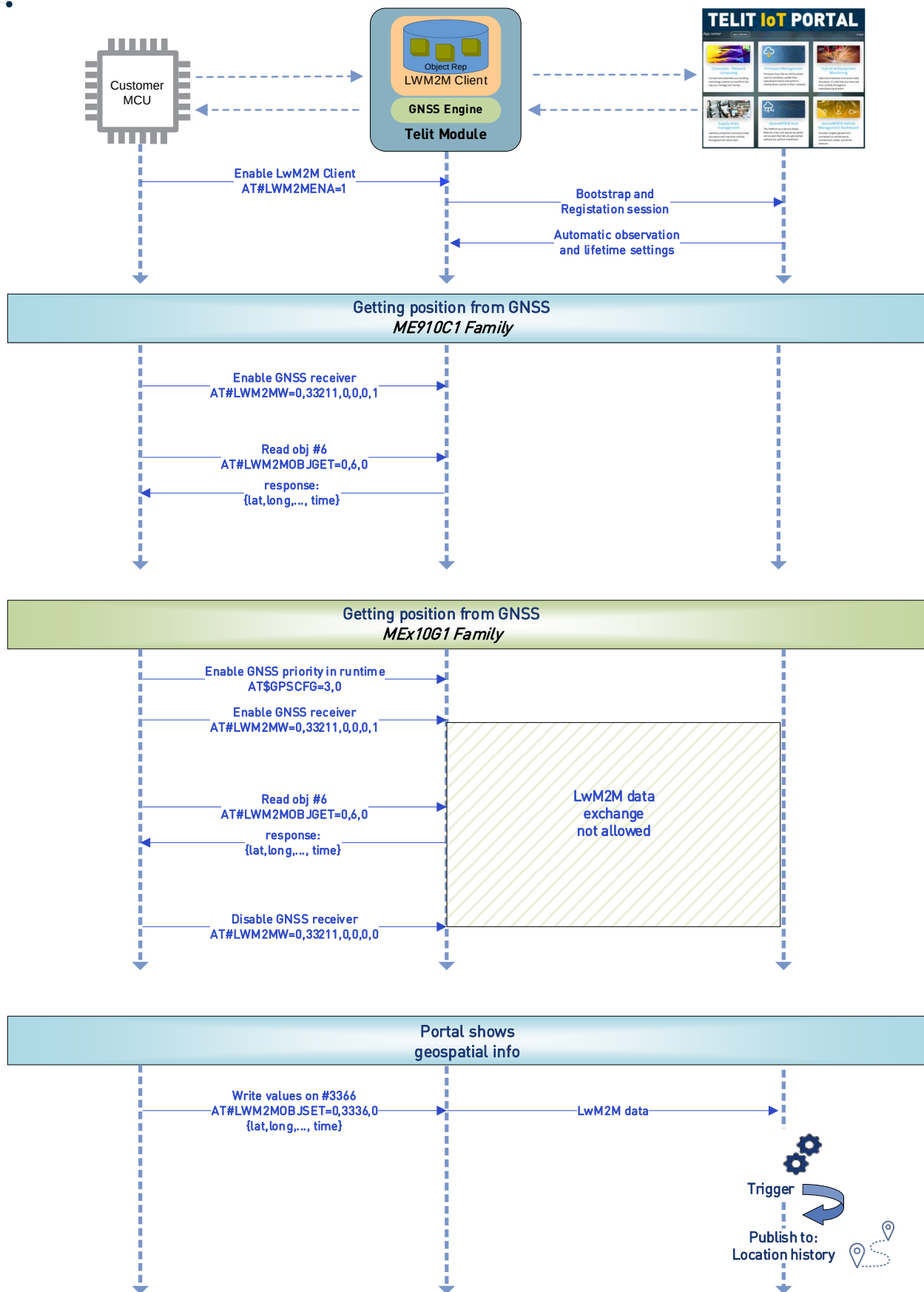


Figure 17: Data flow diagram

4.3. LwM2M data model

According to the requirements defined in the chapter 4.1, the data model has been developed as described in the following paragraphs.

The lifetime is used to configure the time interval settings for the data exchange between LwM2M client and LwM2M server, and it has been set in the Device Profile.

Considering this specific use case, the lifetime parameter does not affect the frequency of the communications and the data consumption.

The command to enable the LwM2M client is sent only once the device is powered on and left enabled as long as the device is running [REQ#1].

4.3.1. OneEdge IoT portal configuration

4.3.1.1. Thing definition

The thing definition must be modified to remove the unneeded fix types on the “Locations” tab, filling the “Fix types to ignore” field with the following keys:

- lwm2m-gps
- lwm2m-network
- network

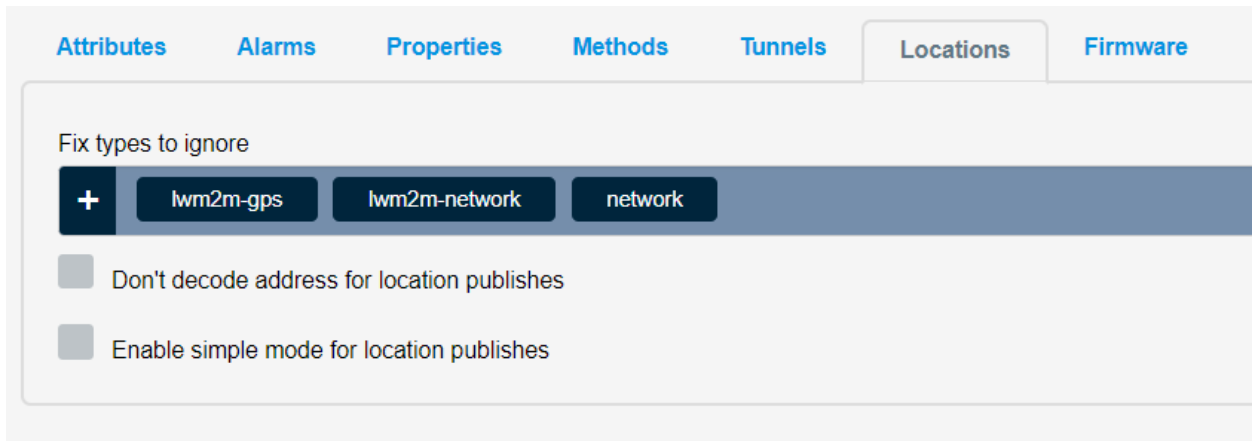


Figure 18: Thing Definition setting to ignore a specific Fix types

4.3.1.2. LwM2M device profile

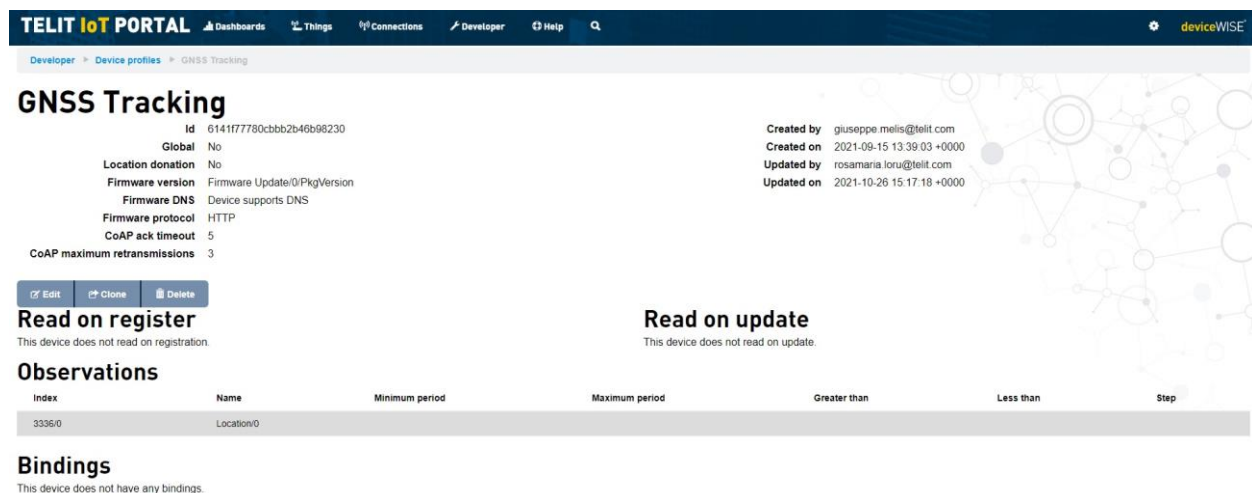
The module is always powered on and the LwM2M client is up and running all the time.

For this reason, the LwM2M client performs full LwM2M registration just once the device is powered on.

According to requirement [REQ#2], since the location data sending must be managed by the host application, the OMA IPSO Location object 3336 is used instead of object 6.

The Lwm2M device profile is customized as below:

- Lifetime = 300 seconds (5 minutes) or higher value
- Observation enabled on object 3336



The screenshot shows the 'GNSS Tracking' device profile in the Telit IoT Portal. The profile details include:

- Id:** 6141f77780cbb2b46b98230
- Global:** No
- Location donation:** No
- Firmware version:** Firmware Update/0/PkgVersion
- Firmware DNS:** Device supports DNS
- Firmware protocol:** HTTP
- CoAP ack timeout:** 5
- CoAP maximum retransmissions:** 3

Metadata on the right:

- Created by:** giuseppe.melis@telit.com
- Created on:** 2021-09-15 13:39:03 +0000
- Updated by:** rosamaria.loru@telit.com
- Updated on:** 2021-10-26 15:17:18 +0000

Buttons: Edit, Clone, Delete.

Read on register: This device does not read on registration.

Read on update: This device does not read on update.

Observations:

Index	Name	Minimum period	Maximum period	Greater than	Less than	Step
3336/0	Location/0					

Bindings: This device does not have any bindings.

Figure 19: Summary of device profile settings

4.3.1.3. Trigger

Thanks to the observation on object 3336, every time the values change in the resources, the Lwm2M client automatically sends the new ones to OneEdge IoT the portal.

According to the requirements [REQ#3] and [REQ#4], the trigger has been developed to publish the thing position on the Portal location map if the tracking device reports a location variation greater than 2 meters.

The trigger is developed as follows:

- Starting point:
 - Event type: lwm2m.instance.change
 - objectId: 3336
 - instanceId: 0
- Action Type 1: location.publish
 - thingKey: \${event.thing.key}
 - lat: \${event.value.6051.value}
 - lng: \${event.value.6052.value}



- fixType: manual
- ts: \${event.value.5518.value}
- debounce: 2

The trigger is showed on Figure 14.

4.3.2. ME910C1 environment

The tracking application starts when the GNSS receiver is enabled by the host application by sending the command `AT#LWM2MW=0,33211,0,0,0,1`.

4.3.3. MEx10G1 environment

Before to enable the GNSS receiver through the command `AT#LWM2MW=0,33211,0,0,0,1`, the host application has to send the command `AT$GPSCFG=3,0` to enable the GNSS priority at runtime.

This step is mandatory to allow the module to use the GNSS engine.

After the tracking process has been completed and the LwM2M client has to send the position data to the LwM2M server, the GNSS receiver must be switched off through the command `AT#LWM2MW=0,33211,0,0,0,0`.

4.3.4. Step by step simulation setup

In the following paragraphs, all steps to simulate the tracking use case are described for both cases: ME910C1 and MEx10G1 environments.

4.3.4.1. Pre-requisites and one-time configuration

The below steps must be considered as preconditions:

- a) Materials: devices, cables, antennas and procedures according to OneEdge Getting Started Guide and GNSS Application Note documents;
- b) Download the files from the following link:
https://github.com/telit/oneedge-projects-resources/tree/main/use-cases/GNSS-tracking/OE_Portal_configuration
- c) Device profile import: load the file "GNSS_Tracking_dev_prof.json" on the OneEdge IoT portal using the section **Developer** → **Device profiles**

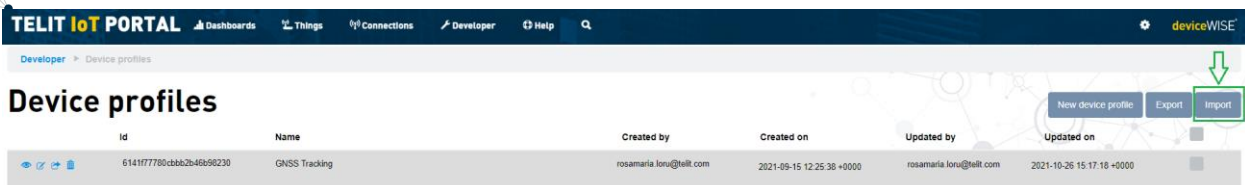


Figure 20: "Device profiles" section with "Import" button in detail

- d) Thing definition import: load the file "GNSS_Tracking_thing_def.json" on the OneEdge IoT portal using the section **Developer** → **Thing definitions**

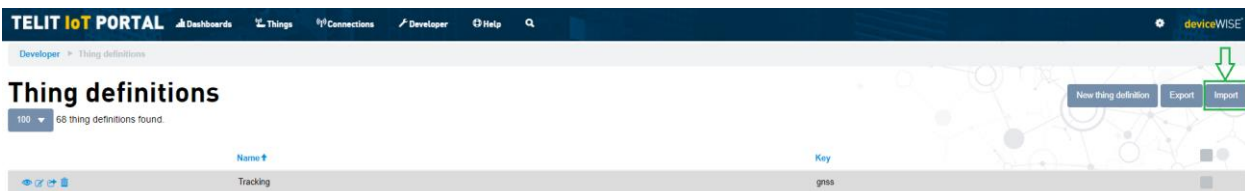


Figure 21: "Thing definitions" section with "Import" button in detail

- e) On Things section of the OneEdge IoT Portal:
- Search your device through the IMEI number
 - Assign the "Tracking" as thing definition
 - Assign the "GNSS Tracking" as device profile

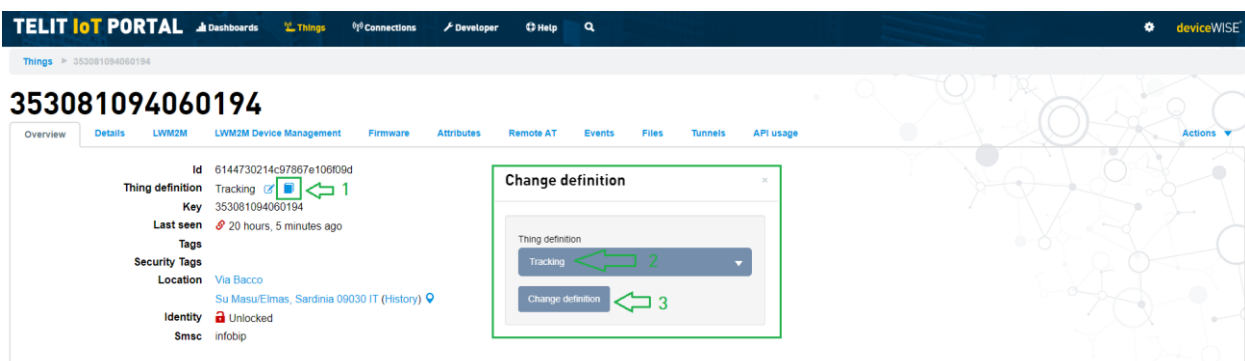


Figure 22: "Change definition" button on thing "Overview" tab

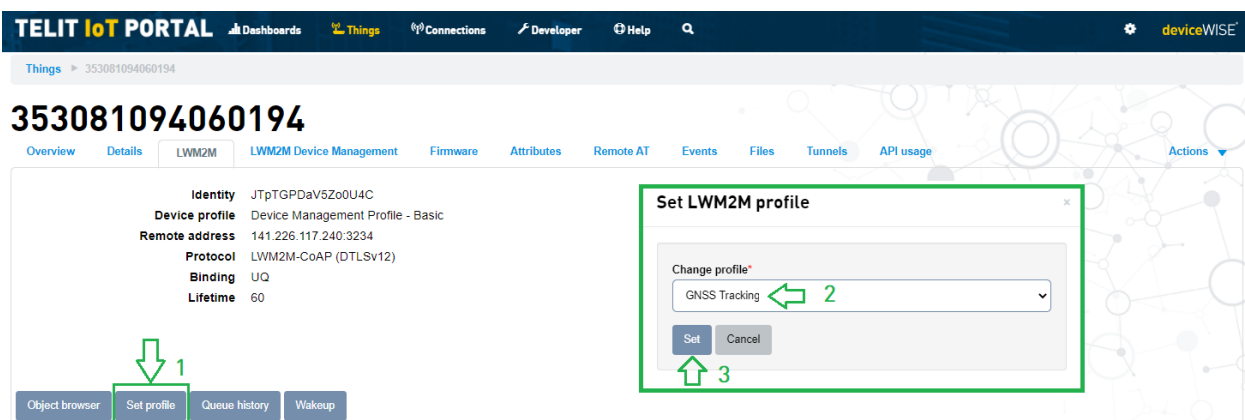


Figure 23: Set profile" button on thing "Lwm2m" tab

- f) IPSO Object file: add the version 2.0 of the object 3336 “object_3336.xml” on section **Developer**→**Object registry** of OneEdge IoT Portal. To do that:
- Press **New object** button to open the edit box popup
 - Copy the content of the XML file and paste it on the edit popup

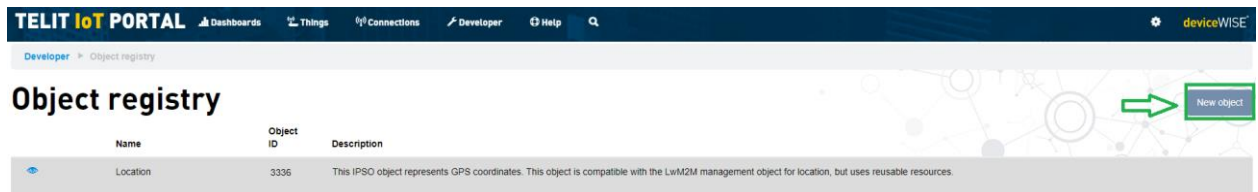


Figure 24: “Object registry” section with “Import” button in detail

- g) GNSS Position to 3336 trigger: import the file “GNSS_tracking_trigger.json” on section **Developer**→**Triggers** of OneEdge IoT Portal (action “1” in Figure 25) and then the trigger has to be started (action “2” in the Figure 25)

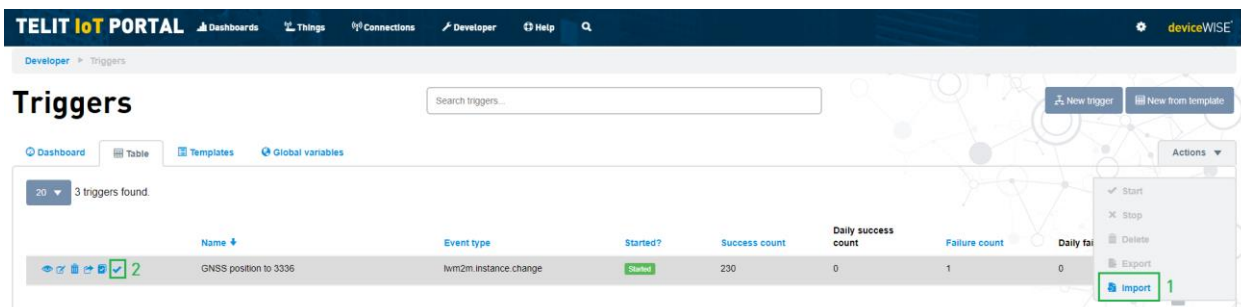


Figure 25: “Triggers” section with “Import” and “Start” buttons in detail

4.3.4.2. AT commands simulation

This paragraph describes the specific step: the host application is responsible to send the AT commands, to manage the URC received on the serial port and also the commands responses.

4.3.4.2.1. Setup of the “tracking” application

This section is related to the LwM2M client setup with all the preconditions. The following steps have to be done only once, at the next power up all these settings are maintained and the procedure could skip directly to paragraph [4.3.4.2.2](#) or [4.3.4.2.3](#).

- a) On OneEdge IoT Portal, configure the device with the “GNSS Tracking” device profile and “Tracking” thing definition.
- b) Load the xml file into the device:

```
AT#M2MWRITE="/XML/object_3336.xml",7765
>>>
```

OK

- c) Reboot the device such that the XML is loaded at the startup:

```
AT#LWM2MSTS=0,999,"coaps://bs.telit.io",1
OK
AT#REBOOT
OK
```

- d) Enable the LwM2M client and wait for registration to the IoT portal:

```
AT#LWM2MW=0,33211,0,0,1,1
OK
AT#LWM2MENA=1
OK
LWM2M-TLT:"BOOTSTRAPPING",SSID=0,"coaps://bs.telit.io"
LWM2M-TLT:"BOOTSTRAPPED",SSID=0,"coaps://bs.telit.io"
LWM2M-TLT:"REGISTERING",SSID=99,"coaps://api-dev.devicewise.com"
LWM2M-TLT:"REGISTERED",SSID=99,"coaps://api-dev.devicewise.com"
```

- e) Create the instance of the custom object

```
AT#LWM2MNEWINST=0,3336,0
OK
```

- f) Restart the LwM2M client to allow the observation to be correctly set:

```
AT#LWM2MENA=0
OK
LWM2M-TLT:"DEREGISTERING",SSID=99,"coaps://api-dev.devicewise.com"
LWM2M-TLT:"DEREGISTERED",SSID=99,"coaps://api-dev.devicewise.com"
LWM2M-TLT:"CLIENT_DISABLED"
```

```
AT#LWM2MENA=1
OK
```

After the registration, automatic actions are performed according to the device profile.

4.3.4.2.2. Sampling loop for ME910C1 environment

At the host application side, the following flow of operations can be developed.

- g) Activate the GNSS receiver:

```
AT#LWM2MW=0,33211,0,0,0,1
OK
```

- h) Read the object 6 values:

```
AT#LWM2MOBJGET=0,6,0
{"bn":"/6/0/",
 "e":[
 {"n":"0","v": 0.000000},
 {"n":"1","v": 0.000000},
 {"n":"2","v": 0.000000},
 {"n":"3","v": 0.000000},
 {"n":"4","sv":"AAAAAA=="}],
```



```
{ "n": "5", "v": 0 },
{ "n": "6", "v": 0.000000 } ]
}
```

- i) Repeat step h until the obtained values are different from 0.

Once the fix is obtained, the result of the `AT#LWM2MOBJGET=0,6,0` command is like to:

```
{ "bn": "/3336/0/",
  "e": [
    { "n": "0", "v": 39.228206 },
    { "n": "1", "v": 9.069231 },
    { "n": "2", "v": 29.000000 },
    { "n": "3", "v": 20.000000 },
    { "n": "4", "sv": "AAAAAA==" },
    { "n": "5", "v": 1633428348 },
    { "n": "6", "v": 0.000000 } ]
}
```

Refer to the paragraph 4.3.4.2.4 for the updated positioning data sending to the LwM2M Server.

4.3.4.2.3. Sampling loop for MEx10G1 environment

At the host application side, the following flow of operations must be performed.

- j) Enable the GNSS priority in runtime

```
AT$GPSCFG=3,0
OK
```

- k) Active the GNSS receiver

```
AT#LWM2MW=0,33211,0,0,0,1
OK
```

- l) Read the values of object 6 with `AT#LWM2MOBJGET=0,6,0` as described in step h of previous paragraph

- m) Repeat the above operation until the obtained values are different from 0.

Once the fix is obtained, the result of the `AT#LWM2MOBJGET=0,6,0` command is like to: see step i of previous paragraph

- n) Disable the GNSS receiver

```
AT#LWM2MW=0,33211,0,0,0,0
OK
```



Warning: steps l and m are in common between ME910C1 and MEx10G1 devices.

4.3.4.2.4. Sending data

According to the observation set on object 3336, the host application must change the values of the resources of this object every time it wants to send a new position to the OneEdge IoT portal.

The sending process is composed by the 2 distinct actions:

- o) Retrieve the location info from object 6

```
AT#LWM2MOBJGET=0,6,0
{"bn":"/6/0/",
 "e":[
  {"n":"0","v":39.228206},
  {"n":"1","v":9.069231},
  {"n":"2","v":29.000000},
  {"n":"3","v":20.000000},
  {"n":"4","sv":"AAAAAA=="},
  {"n":"5","v":1633428348},
  {"n":"6","v":0.000000}]
}
```

- p) Write the values of the object 6 into the object 3336

```
AT#LWM2MOBJSET=0,3336,0
{"bn":"/3336/0/",
 "e":[
  {"n":"6051","v":39.228206},
  {"n":"6052","v":9.069231},
  {"n":"6053","v":20.0},
  {"n":"5518","v":1633428348},
  {"n":"5517","sv":"AAAAAA=="}]
}<Ctrl-Z>
```

4.3.4.3. Portal Publish

Every time the OneEdge IoT portal receives new object 3336 resources values due to the operation in step **p**, the trigger is fired and a new point on the map is published, as in Figure 26.

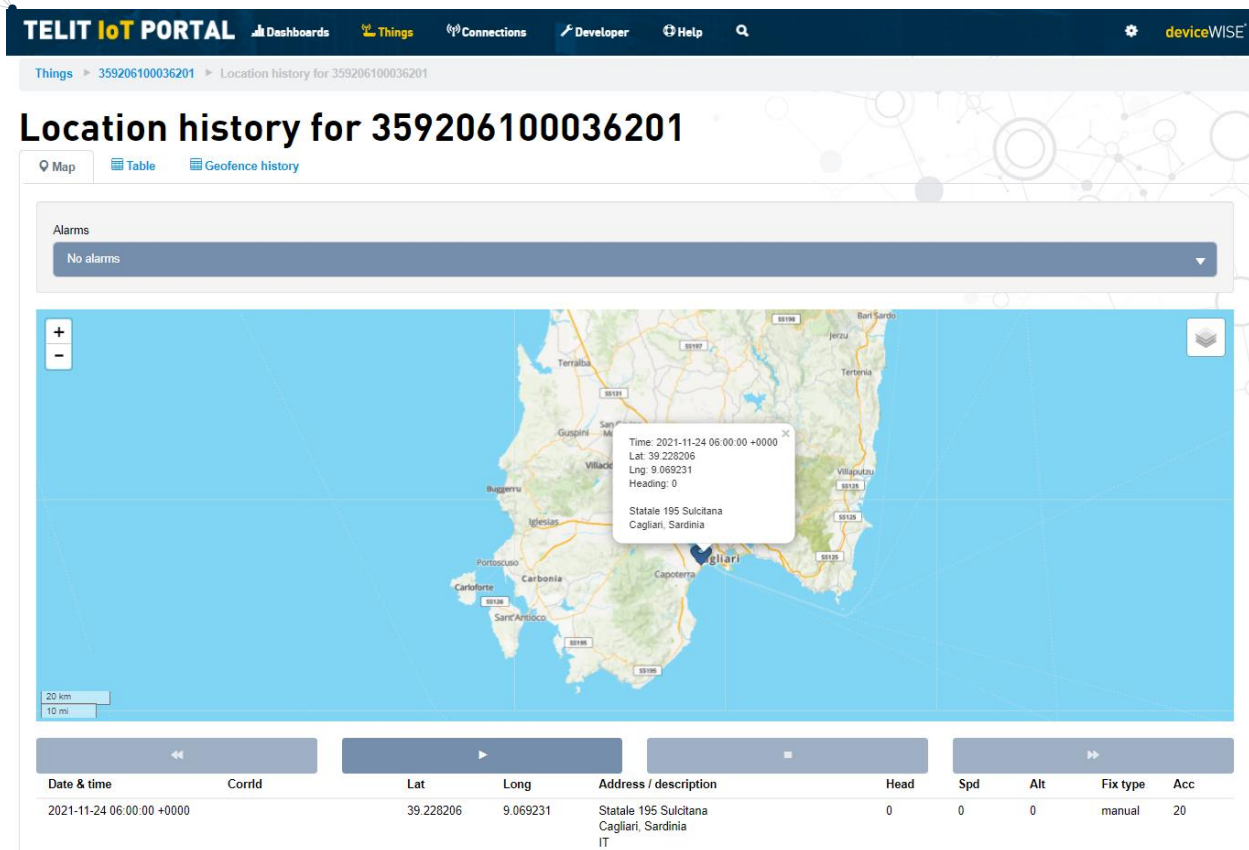


Figure 26: Sa Illetta campus position

4.3.4.4. Python script simulation

A python script has been developed to simulate the basic actions that a host application should do to manage a tracking application.

The Python script sends AT commands and handles the responses coming from the LwM2M client, according to the flow described in paragraph [4.3.4.2](#).

The Figures 27 and 28 below show the result of a real run of the tracking application: the tracked device was moving westward and published 13 points on the map over a 26 minutes timespan.

The last point in the map is identified by a “balloon” tip.

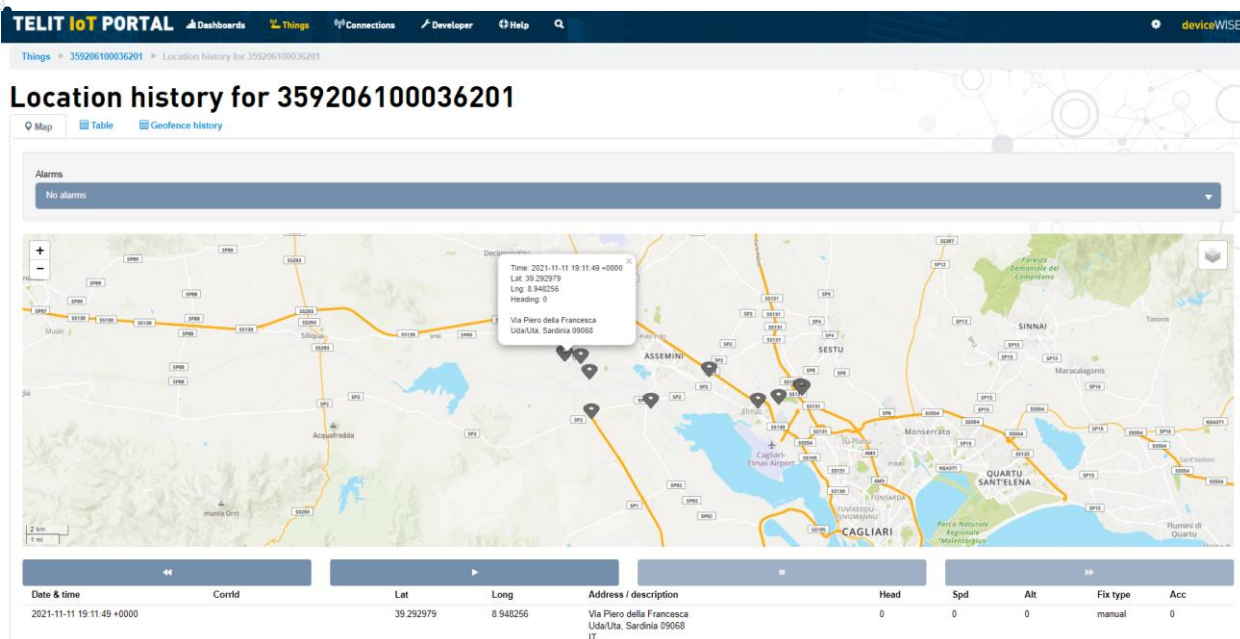


Figure 27: Map view – car tracking

TELIT IoT PORTAL

Dashboard

Things

Connections

Developer

Help

Q

DE-AE-EMEA

deviceWISE

Things

>

359206100036201

>

Location history for 359206100036201

Location history for 359206100036201

Map

Table

Geofence history

Download CSV

Date & time	Corrid	Latitude	Longitude	Delta	Address / description	Heading	Speed	Altitude	Fix type	Accuracy	State	Country
2021-11-11 18:47:51 +0000	39.278331	9.078722	26 meters	Km 7+650 ex SS131 Sestu Sardinia 09028 IT	0.0	0	0.0 meters	manual	0.0 meters	Sardinia	IT	
2021-11-11 18:49:47 +0000	39.278394	9.078331	34 meters	Km 7+650 ex SS131 Sestu Sardinia 09028 IT	0.0	0	0.0 meters	manual	0.0 meters	Sardinia	IT	
2021-11-11 18:51:47 +0000	39.278708	9.077697	64 meters	Km 7+650 ex SS131 Sestu Sardinia 09028 IT	0.0	0	0.0 meters	manual	0.0 meters	Sardinia	IT	
2021-11-11 18:53:48 +0000	39.274322	9.065848	1131 meters	SP8 Su Masu/Elmas Sardinia 09030 IT	0.0	0	0.0 meters	manual	0.0 meters	Sardinia	IT	
2021-11-11 18:55:48 +0000	39.272705	9.054570	968 meters	SS130 Su Masu/Elmas Sardinia 09030 IT	0.0	0	0.0 meters	manual	0.0 meters	Sardinia	IT	
2021-11-11 18:57:48 +0000	39.285939	9.027791	2737 meters	Pedemontana Assèmini/Assèmini Sardinia 09032 IT	0.0	0	0.0 meters	manual	0.0 meters	Sardinia	IT	
2021-11-11 18:59:48 +0000	39.272725	8.995708	3131 meters	Pedemontana Assèmini/Assèmini Sardinia 09032 IT	0.0	0	0.0 meters	manual	0.0 meters	Sardinia	IT	
2021-11-11 19:01:48 +0000	39.268231	8.963280	2638 meters	unnamed road Uda/Uta Sardinia 09068 IT	0.0	0	0.0 meters	manual	0.0 meters	Sardinia	IT	
2021-11-11 19:03:48 +0000	39.284996	8.962199	1868 meters	unnamed road Uda/Uta Sardinia 09068 IT	0.0	0	0.0 meters	manual	0.0 meters	Sardinia	IT	
2021-11-11 19:05:48 +0000	39.291410	8.957151	835 meters	76A Via Regina Margherita Uda/Uta Sardinia 09010 IT	0.0	0	0.0 meters	manual	0.0 meters	Sardinia	IT	
2021-11-11 19:07:48 +0000	39.292627	8.948577	750 meters	Via Piero della Francesca Uda/Uta Sardinia 09068 IT	0.0	0	0.0 meters	manual	0.0 meters	Sardinia	IT	
2021-11-11 19:09:48 +0000	39.292772	8.948455	19 meters	Via Piero della Francesca Uda/Uta Sardinia 09068 IT	0.0	0	0.0 meters	manual	0.0 meters	Sardinia	IT	
2021-11-11 19:11:49 +0000	39.252979	8.948256	28 meters	Via Piero della Francesca Uda/Uta Sardinia 09068 IT	0.0	0	0.0 meters	manual	0.0 meters	Sardinia	IT	

Figure 28: Table view – car tracking

Below the messages sequence is reported.

4.3.4.4.1. ME910C1 Environment

This example shows how the module finds the fix and sends the first two positions to the portal.

```
USE DEVICE PROFILE WITH OBSERVE 3336/0
```

```
USE A TRIGGER TO CONVERT THE 3336 RESOURCE INTO A MAP POSITION
```

```
sending: AT+LWM2MW=0,33211,0,0,0,1
```

```
OK
```

START LOOP for fix

waiting for fix

sending: AT#LWM2MOBJGET=0,6,0

response:{"n":"/6/0/","e":[{"n":"0","v":0.000000}, {"n":"1","v":0.000000}, {"n":"2","v":0.000000}, {"n":"3","v":0.000000}, {"n":"4","sv":"AAAAAA=="}, {"n":"5","v":0}, {"n":"6","v":0.000000}]}

latitude from GNSS is 0.0

longitude from GNSS is 0.0

altitude from GNSS is 0.0

velocity from GNSS is AAAAAA==

timestamp from GNSS is 0

speed from GNSS is 0.0

waiting for fix

sending: AT#LWM2MOBJGET=0,6,0

...

//these attempts go ahead until the response of the AT#LWM2MOBJGET=0,6,0 will be different from zero

...

waiting for fix

sending: AT#LWM2MOBJGET=0,6,0

response:{"n":"/6/0/","e":[{"n":"0","v":39.228140}, {"n":"1","v":9.069387}, {"n":"2","v":64.000000}, {"n":"3","v":35.000000}, {"n":"4","sv":"AAAAAA=="}, {"n":"5","v":1633623681}, {"n":"6","v":0.080000}]}

latitude from GNSS is 39.22814

longitude from GNSS is 9.069387

altitude from GNSS is 64.0

velocity from GNSS is AAAAAA==

timestamp from GNSS is 1633623681

speed from GNSS is 0.08

fix found

START LOOP for data upload

LOOP 1

sending: AT#LWM2MOBJGET=0,6,0

response:{"n":"/6/0/","e":[{"n":"0","v":39.228159}, {"n":"1","v":9.069274}, {"n":"2","v":35.000000}, {"n":"3","v":3.000000}, {"n":"4","sv":"AAAAAA=="}, {"n":"5","v":1633623684}, {"n":"6","v":0.000000}]}

latitude from GNSS is 39.228159

longitude from GNSS is 9.069274

```

altitude from GNSS is 35.0

velocity from GNSS is AAAAAA==

timestamp from GNSS is 1633623684

speed from GNSS is 0.0

sending: AT#LWM2MOBJSET=0,3336,0

sending: {"bn":"/3336/0/", "e":[{"n":"6051", "v":39.228159}, {"n":"6052", "v":9.069274}, {"n":"6053", "v":3.0}, {"n":"5518", "v":1633623684}, {"n":"5517", "sv":"AAAAAA=="}]}

received: {"bn":"/3336/0/", "e":[{"n":"6051", "v":39.228159}, {"n":"6052", "v":9.069274}, {"n":"6053", "v":3.0}, {"n":"5518", "v":1633623684}, {"n":"5517", "sv":"AAAAAA=="}]}\r\x1a\r\nOK\r\n'

LOOP 2

sending: AT#LWM2MOBJGET=0,6,0

{"n":"/6/0/", "e":[{"n":"0", "v":39.226982}, {"n":"1", "v":9.071212}, {"n":"2", "v":0.000000}, {"n":"3", "v":3.000000}, {"n":"4", "sv":"AIYFAA=="}, {"n":"5", "v":1633624281}, {"n":"6", "v":5.450615}]}OK

response: {"n":"/6/0/", "e":[{"n":"0", "v":39.226982}, {"n":"1", "v":9.071212}, {"n":"2", "v":0.000000}, {"n":"3", "v":3.000000}, {"n":"4", "sv":"AIYFAA=="}, {"n":"5", "v":1633624281}, {"n":"6", "v":5.450615}]}

latitude from GNSS is 39.226982

longitude from GNSS is 9.071212

altitude from GNSS is 0.0

velocity from GNSS is AIYFAA==

timestamp from GNSS is 1633624281

speed from GNSS is 5.450615

sending: AT#LWM2MOBJSET=0,3336,0

sending: {"bn":"/3336/0/", "e":[{"n":"6051", "v":39.226982}, {"n":"6052", "v":9.071212}, {"n":"6053", "v":3.0}, {"n":"5518", "v":1633624281}, {"n":"5517", "sv":"AIYFAA=="}]}

received: {"bn":"/3336/0/", "e":[{"n":"6051", "v":39.226982}, {"n":"6052", "v":9.071212}, {"n":"6053", "v":3.0}, {"n":"5518", "v":1633624281}, {"n":"5517", "sv":"AIYFAA=="}]}\r\x1a\r\nOK\r\n'

```

According to paragraph [“3.2.5. Position management on OneEdge IoT portal”](#), every 3336 object modification through AT#LWM2MOBJSET produced a LwM2M notify sent to the LwM2M server and then a Location point published on the thing map through the trigger.

4.3.4.4.2. MEx10G1 Environment

Since the procedure to manage the GNSS data on MEx10G1 devices is a bit different than ME910C1, the Python script has been modified according to paragraph [4.3.4.2.3](#).

The report below shows the output of the script in this case.

```
USE DEVICE PROFILE WITH OBSERVE 3336/0

USE A TRIGGER TO CONVERT THE 3336 RESOURCE INTO A MAP POSITION

sending: AT$GPSCFG=3,0

OK

sending: AT#LWM2MW=0,33211,0,0,0,1

OK

GNSS Session is ongoing


START LOOP for fix

waiting for fix

sending: AT#LWM2MOBJGET=0,6,0

response:{"n":"/6/0/", "e":[{"n":"0", "v":0.000000}, {"n":"1", "v":0.000000}, {"n":"2", "v":0.000000}, {"n":"3", "v":0.000000}, {"n":"4", "sv":"AAAAAA=="}, {"n":"5", "v":0}, {"n":"6", "v":0.000000}]}

latitude from GNSS is 0.0

longitude from GNSS is 0.0

altitude from GNSS is 0.0

velocity from GNSS is AAAAAA==

timestamp from GNSS is 0

speed from GNSS is 0.0

...

//these attempts go ahead until the response of the AT#LWM2MOBJGET=0,6,0 will be different from zero

...

waiting for fix

sending: AT#LWM2MOBJGET=0,6,0

{"n":"/6/0/", "e":[{"n":"0", "v":39.228239}, {"n":"1", "v":9.069245}, {"n":"2", "v":12.744839}, {"n":"3", "v":3.535534}, {"n":"4", "sv":"AAAAAA=="}, {"n":"5", "v":1635519968}, {"n":"6", "v":0.000000}]}OK

response:{"n":"/6/0/", "e":[{"n":"0", "v":39.228239}, {"n":"1", "v":9.069245}, {"n":"2", "v":12.744839}, {"n":"3", "v":3.535534}, {"n":"4", "sv":"AAAAAA=="}, {"n":"5", "v":1635519968}, {"n":"6", "v":0.000000}]}

latitude from GNSS is 39.228239

longitude from GNSS is 9.069245

[2altitude from GNSS is 12.744839
```

velocity from GNSS is AAAAAA==

timestamp from GNSS is 1635519968

speed from GNSS is 0.0

fix found

sending: AT#LWM2MW=0,33211,0,0,0,0

OK

WWAN Session is ongoing

Position sent through SET command

sending: AT#LWM2MOBJGET=0,6,0

{"n":"/6/0/","e":[{"n":"0","v":39.228254},{"n":"1","v":9.069219},{"n":"2","v":6.829727},{"n":"3","v":4.300464},{"n":"4","sv":"AAAAAA=="},{"n":"5","v":1635520274},{"n":"6","v":0.083037}]}OK

response:{"n":"/6/0/","e":[{"n":"0","v":39.228254},{"n":"1","v":9.069219},{"n":"2","v":6.829727},{"n":"3","v":4.300464},{"n":"4","sv":"AAAAAA=="},{"n":"5","v":1635520274},{"n":"6","v":0.083037}]}

latitude from GNSS is 39.228254

longitude from GNSS is 9.069219

altitude from GNSS is 6.829727

velocity from GNSS is AAAAAA==

timestamp from GNSS is 1635520274

speed from GNSS is 0.083037

sending: AT#LWM2MOBJSET=0,3336,0

sending: {"bn":"/3336/0/","e":[{"n":"6051","v":39.228254},{"n":"6052","v":9.069219},{"n":"6053","v":4.300464},{"n":"5518","v":1635520274},{"n":"5517","sv":"AAAAAA=="}]}

received: {"bn":"/3336/0/","e":[{"n":"6051","v":39.228254},{"n":"6052","v":9.069219},{"n":"6053","v":4.300464},{"n":"5518","v":1635520274},{"n":"5517","sv":"AAAAAA=="}]}\r\x1a\r\nOK\r\n'

START LOOP for data upload

LOOP 1

sending: AT#LWM2MW=0,33211,0,0,0,1

OK

GNSS Session is ongoing

sending: AT#LWM2MOBJGET=0,6,0

response:{"n":"/6/0/","e":[{"n":"0","v":39.228256},{"n":"1","v":9.069249},{"n":"2","v":18.225227},{"n":"3","v":3.535534},{"n":"4","sv":"AAAAAA=="},{"n":"5","v":1635520400},{"n":"6","v":0.000000}]}

latitude from GNSS is 39.228256

longitude from GNSS is 9.069249

altitude from GNSS is 18.225227

velocity from GNSS is AAAAAA==

timestamp from GNSS is 1635520400

speed from GNSS is 0.0

sending: AT#LWM2MW=0,33211,0,0,0,0

OK

WWAN Session is ongoing

sending: AT#LWM2MOBJSET=0,3336,0

sending: {"bn":"/3336/0/","e":[{"n":"6051","v":39.228256},{"n":"6052","v":9.069249},{"n":"6053","v":3.535534},{"n":"5518","v":1635520400},{"n":"5517","sv":"AAAAAA=="}]}

received: {"bn":"/3336/0/","e":[{"n":"6051","v":39.228256},{"n":"6052","v":9.069249},{"n":"6053","v":3.535534},{"n":"5518","v":1635520400},{"n":"5517","sv":"AAAAAA=="}]}\r\x1a\r\nOK\r\n'

5. POSITIONING USE CASE

Another scope covered by GNSS applications is determining the position of an object in space.

Some examples could be:

- **Parcels tracking:** the real time position can be known at specific transit points or periodical scheduled checks;
- **Military/law enforcement applications:** home detention monitoring.
- **Vehicle rental:** at specific interval times the deployment of the fleet can be mapped.
- **Animal collars:** GNSS-based applications are used to support research institutes to know on a regular schedule the exact position of wild animals in nature reserves.

This use case is particularly suitable for those applications where accurate tracking is not required and which have limited battery capabilities.

5.1. High-level requirements of the application

The positioning device sends geo-spatial information to the OneEdge IoT portal with low frequency, for example once a day.

The application uses the GNSS internal receiver of the Telit module, for this reason the IPSO location object 3336 and also the trigger is not necessary.

Below the main requirements/features:

5.1.1. Power Supply

- **[REQ #1] The device is battery powered**

Since the positioning application is not always running, it does not require electrical power supply or similar.

5.1.2. Device to Portal direction

- **[REQ #2] The positioning device sends geo-spatial information from internal GNSS receiver to the OneEdge IoT portal once a day**

The host application manages a default sending rate.

Geo-spatial information are the following:



- ✓ Latitude
- ✓ Longitude
- ✓ Altitude
- ✓ Radius/Accuracy
- ✓ Velocity/Heading
- ✓ Timestamp
- ✓ Speed

5.1.3. Portal view

- **[REQ #3] Map**

The geo-spatial information are plotted on a map on the OneEdge IoT portal.

- **[REQ #4] Table**

The geo-spatial information are available in text format as a list on the OneEdge IoT Portal or as a downloadable CSV file.

5.2. Data flow diagram

The image below shows the flow of AT commands, LwM2M messages and events across the actors involved in the positioning use case.

The vertical arrows indicate the time, starting from the top to the bottom.

All the four features are represented:

- getting position from GNSS engine (for both families);
- the registration phase;
- sending position to the portal to shows the geo-spatial information;
- the de-registration phase.

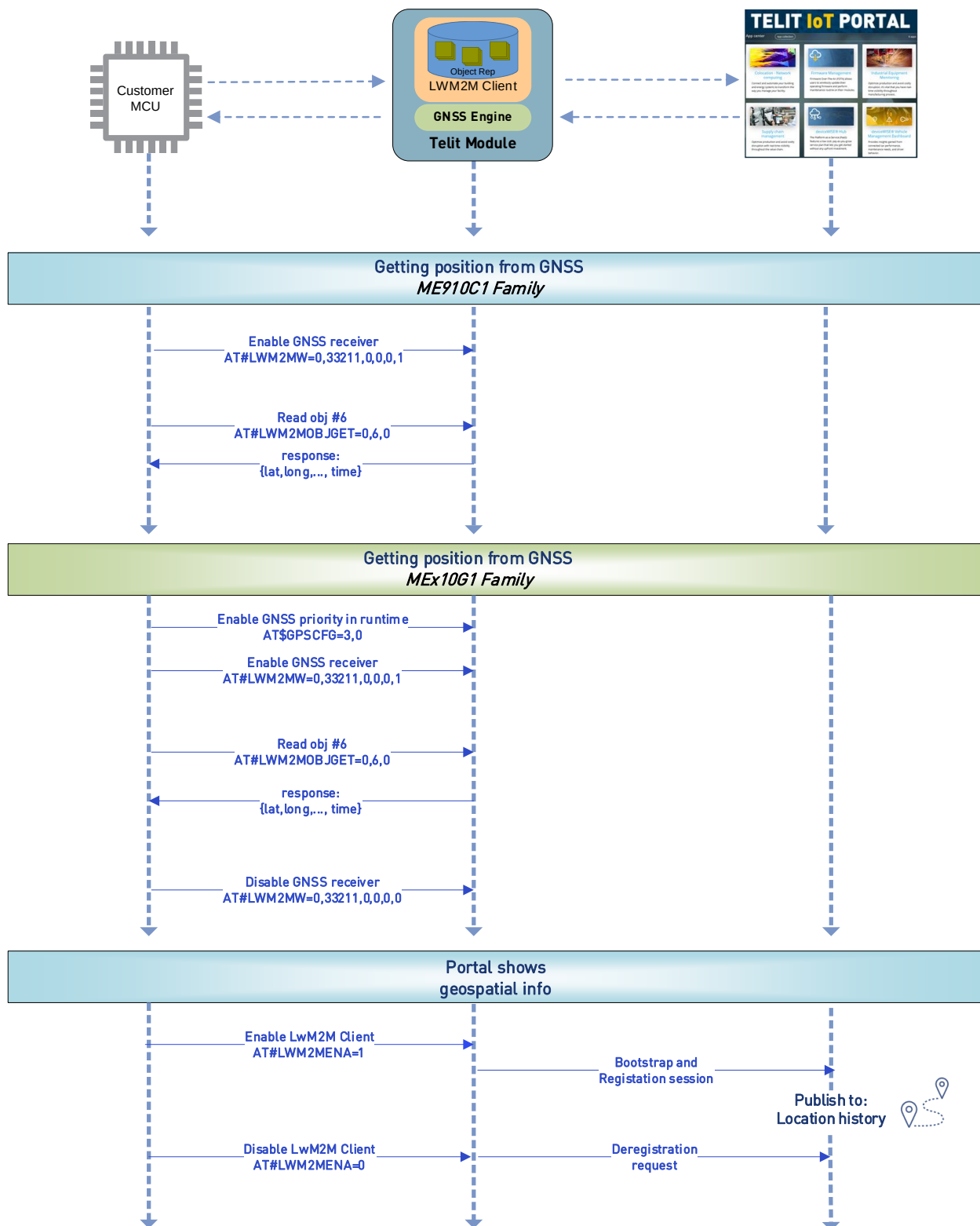


Figure 29: Data flow diagram

5.3. LwM2M data model

According to requirements exposed in the [chapter 5.1](#) the data model has been developed as described in the following paragraphs.

The command to enable the LwM2M client is sent after the device obtained the fix and the connection between LwM2M client and server is killed once the data exchange is completed [REQ#1].

5.3.1. OneEdge IoT portal configuration

5.3.1.1. Thing definition

The thing definition has been maintained equal to the previous use case, except for the “fix type to ignore” settings: the fix type “lwm2m-gps” is a mandatory parameter to handle the position through OMA object 6.

The thing definition “Positioning” is configured as reported in Figure 30.



Figure 30: Thing Definition setting to ignore a specific Fix types

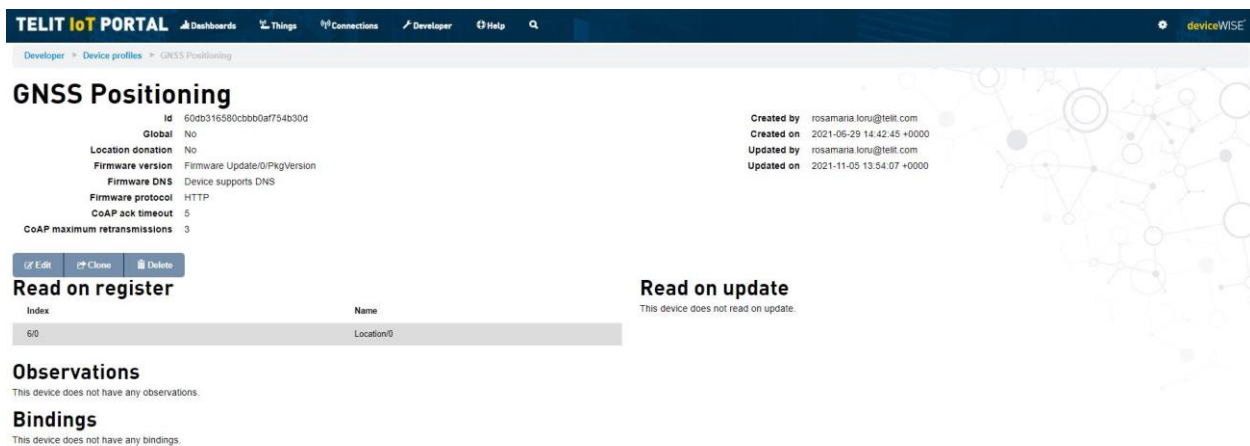
5.3.1.2. LwM2M device profile

The module is always powered on but the LwM2M client performs the full LwM2M registration only after the fix has been obtained and the resources of the object 6 contain the positioning data.

According to the device profile settings, the positioning information are sent to the LwM2M server during the client registration phase.

The LwM2M device profile is customized as below:

- Lifetime = 300 seconds (5 minutes) or higher value
- Read on register for object 6



TELIT IoT PORTAL | Dashboards | Things | Connections | Developer | Help

Developer > Device profiles > GNSS Positioning

GNSS Positioning

Id: 60db316580cbb0af754b30d
 Global: No
 Location donation: No
 Firmware version: Firmware Update/0/PkgVersion
 Firmware DNS: Device supports DNS
 Firmware protocol: HTTP
 CoAP ack timeout: 5
 CoAP maximum retransmissions: 3

Created by: rosamaria Ionu@telit.com
 Created on: 2021-06-29 14:42:45 +0000
 Updated by: rosamaria Ionu@telit.com
 Updated on: 2021-11-05 13:54:07 +0000

[Edit](#) [Clone](#) [Delete](#)

Read on register

Index	Name
6/0	Location/0

Read on update

This device does not read on update.

Observations

This device does not have any observations.

Bindings

This device does not have any bindings.

Figure 31: Summary of device profile settings

The OneEdge IoT portal is configured to automatically publish the data present in the OMA object 6 in the Location history section of the thing overview page.

The values of the object 6 resources are represented as points on the map or a row in the Location table.

5.3.2. ME910C1 environment

The positioning application starts when the GNSS receiver is enabled by the host application, sending the command `AT#LWM2MW=0,33211,0,0,0,1`.

5.3.3. MEx10G1 environment

Before to enable the GNSS receiver through the command `AT#LWM2MW=0,33211,0,0,0,1`, the host application has to send the command `AT$GPSCFG=3,0` to enable the GNSS priority at runtime.

This step is mandatory to allow the module to use the GNSS engine.

After the positioning process has been completed and the LwM2M client has to send the position data to the LwM2M server, the GNSS receiver must be switched off through the command `AT#LWM2MW=0,33211,0,0,0,0`.

5.3.4. Step by step simulation setup

In the following paragraphs, all steps to simulate the positioning use case are described for both ME910C1 and MEx10G1 environments.

5.3.4.1. Pre-requisites and one-time configuration

The below steps must be considered as preconditions:

a) Materials: devices, cables, antennas and procedures according to OneEdge Getting Started Guide and GNSS Application Note documents;

b) Download the files from the following link:

https://github.com/telit/oneedge-projects-resources/tree/main/use-cases/GNSS-tracking/OE_Portal_configuration

c) Device profile import: load the file “GNSS_Positioning_dev_prof.json” on the OneEdge IoT portal using the section **Developer** → **Device profiles**

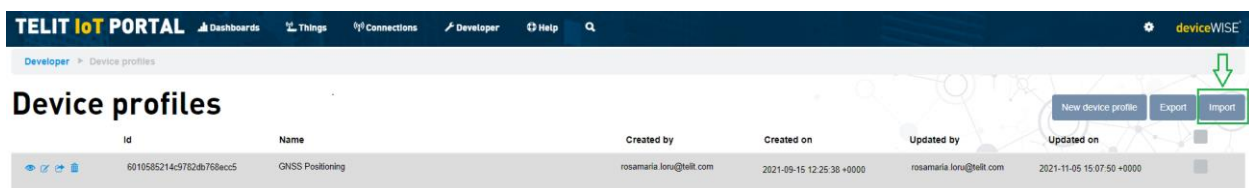


Figure 32: “Device profiles” section with “Import” button in detail

d) Thing definition import: load the file “GNSS_Positioning_thing_def.json” on the OneEdge IoT portal using the section **Developer** → **Thing definitions**

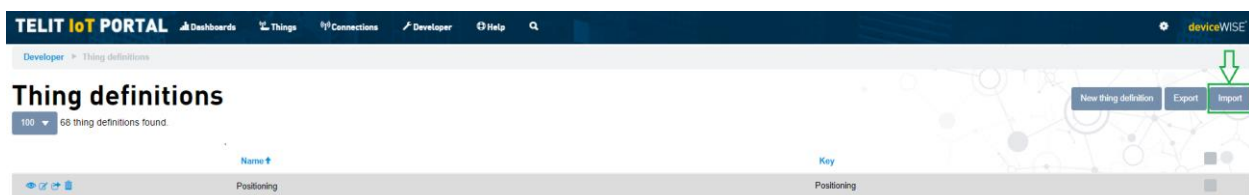


Figure 33: “Thing definitions” section with “Import” button in detail

- e) On Things section of the OneEdge IoT Portal:
- Search your device through the IMEI number
 - Assign the “Positioning” as thing definition
 - Assign the “GNSS Positioning” as device profile

5.3.4.2. AT commands simulation

This paragraph describes the specific steps: the host application is responsible to send the AT commands, to manage the URC received on the serial port and also the commands responses.

5.3.4.2.1. Setup of the “Positioning” application

This section is related to the LwM2M client setup with all the preconditions. The following steps have to be done only once, at the next power up all these settings are maintained and the procedure could skip directly to the following paragraphs (step h and later).

- f) On OneEdge IoT Portal, configure the device with the “GNSS Positioning” device profile and “Positioning” thing definition.
- g) Reset the LwM2M client and reboot the device:

```
AT#LWM2MSTS=0,999,"coaps://bs.telit.io",1
OK
AT#REBOOT
OK
```

5.3.4.2.2. Sampling loop for ME910C1 environment

At the host application side the following flow of operations can be developed.

- h) Activate the GNSS receiver:

```
AT#LWM2MW=0,33211,0,0,0,1
OK
```

- i) Read the object 6 values:

```
AT#LWM2MOBJGET=0,6,0
{"bn":"/6/0/",
 "e":[
  {"n":"0","v": 0.000000},
  {"n":"1","v": 0.000000},
  {"n":"2","v": 0.000000},
  {"n":"3","v": 0.000000},
  {"n":"4","sv":"AAAAAA=="},
  {"n":"5","v":0},
  {"n":"6","v": 0.000000}]
}
```

- j) Repeat the step i until the obtained values are different from 0.

Once the fix is obtained, the result of the AT#LWM2MOBJGET=0,6,0 command is like to:

```
{"bn":"/6/0/",
 "e":[
  {"n":"0","v":39.228206},
  {"n":"1","v":9.069231},
  {"n":"2","v":29.000000},
  {"n":"3","v":20.000000},
  {"n":"4","sv":"AAAAAA=="},
  {"n":"5","v":1633428348},
  {"n":"6","v":0.000000}]
}
```

Refer to the paragraph [5.3.4.2.4](#) for the updated positioning data sending to the LwM2M Server.

5.3.4.2.3. Sampling loop for MEx10G1 environment

At the host application side the following flow of operations must be performed

- k) Enable the GNSS priority in runtime:

```
AT$GPSCFG=3,0
OK
```

- l) Active the GNSS receiver:

```
AT#LWM2MW=0,33211,0,0,0,1
OK
```

- m) Read the values of object 6 with `AT#LWM2MOBJGET=0,6,0` as described in step i of previous paragraph
- n) Repeat the above operation until the obtained values are different from 0.

Once the fix is obtained, the result of the `AT#LWM2MOBJGET=0,6,0` command is like to: see step j of previous paragraph

- o) Disable the GNSS receiver

```
AT#LWM2MW=0,33211,0,0,0,0
OK
```



Warning: steps **m** and **n** are in common between ME910C1 and MEx10G1 devices.

5.3.4.2.4. Sending data and Portal publish

According to the “read on register” set on object 6, the host application must enable the Lwm2M client every time it wants to send a new position to the OneEdge IoT portal.

The process starts when the host application sends the AT command `AT#LWM2MENA=1`.

- p) Enable the URC messages:

```
AT#LWM2MW=0,33211,0,0,1,1
OK
```

- q) Enable the Lwm2M client and wait for registration to the IoT portal

```
AT#LWM2MENA=1
OK
LWM2M-TLT: "BOOTSTRAPPING",SSID=0,"coaps://bs.telit.io"
LWM2M-TLT: "BOOTSTRAPPED",SSID=0,"coaps://bs.telit.io"
LWM2M-TLT: "REGISTERING",SSID=99,"coaps://api-dev.devicewise.com"
LWM2M-TLT: "REGISTERED",SSID=99,"coaps://api-dev.devicewise.com"
```

The values of this object 6 are sent to the LwM2M server during the client registration phase and put in the Location history portal section for the position publishing.

5.3.4.3. LwM2M Client Switch off

- r) Disable the LwM2M client to save battery charge

```
AT#LWM2MENA=0
OK
LWM2M-TLT: "DEREGISTERING", SSID=99, "coaps://api-dev.devicewise.com"
LWM2M-TLT: "DEREGISTERED", SSID=99, "coaps://api-dev.devicewise.com"
LWM2M-TLT: "CLIENT_DISABLED"
```

5.3.4.4. Python script simulation

A python script has been developed to simulate the basic actions that an host application should do to manage a positioning application.

The Python script sends AT commands and handles the responses coming from the LwM2M client according to the flow described in paragraph 5.3.4.2.

The Figure 34 below shows the result of a real run of the positioning application: the positioning device was moving across Sardinia island and published 4 points on the map during 4 days.

The last point in the map is identified by a “balloon” tip.

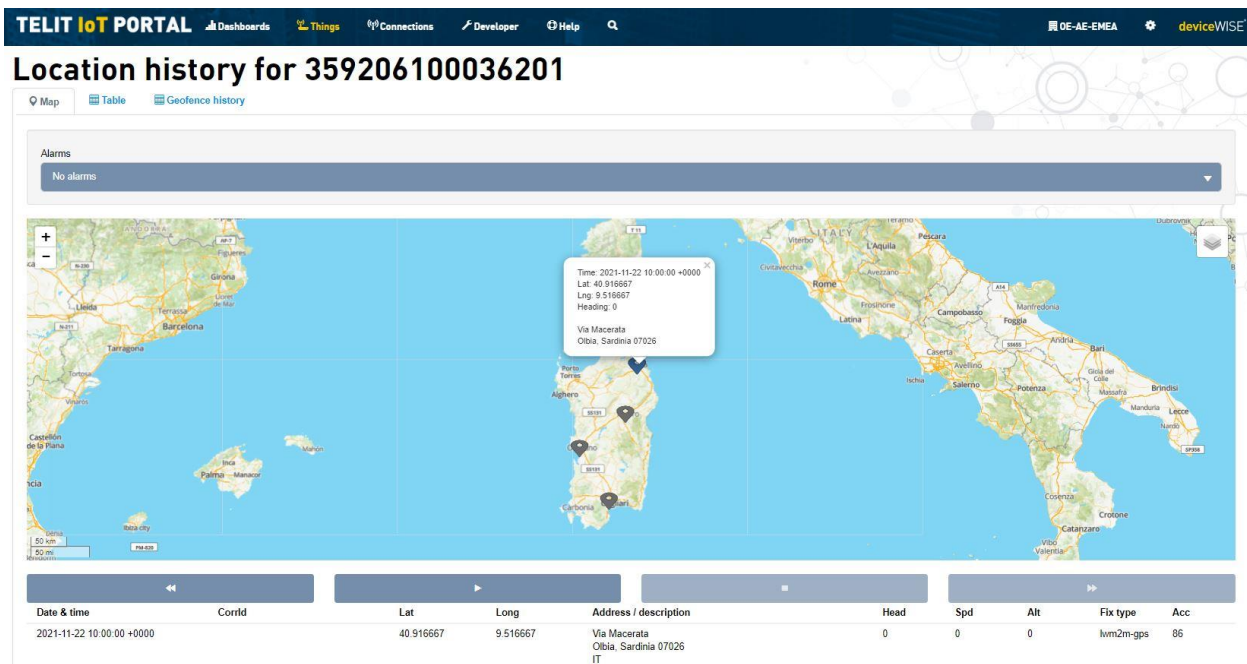


Figure 34: Map view – positioning

Below the sequence of messages is reported.

5.3.4.4.1. ME910C1 Environment

This example shows how the module find the fix and send the first two positions of the module to the portal.

```
OPEN THE SERIAL PORT

USE DEVICE PROFILE WITH READ ON REGISTER ON OBJ 6/0

START LOOP for data upload

LOOP 1

sending: AT#LWM2MW=0,33211,0,0,0,1

OK


START LOOP for fix

waiting for fix

sending: AT#LWM2MR=0,3,0,13,0

response:#LWM2MR: 1637234650

sending: AT#LWM2MOBJGET=0,6,0

response:{"n":"/6/0/","e":[{"n":"0","v":0.000000}, {"n":"1","v":0.000000}, {"n":"2","v":0.000000}, {"n":"3","v":0.000000}, {"n":"4","sv":"AAAAAA=="}, {"n":"5","v":0}, {"n":"6","v":0.000000}]}

latitude from GNSS is 0.0

longitude from GNSS is 0.0

altitude from GNSS is 0.0

radius from GNSS is 0.0

velocity from GNSS is AAAAAA==

timestamp from GNSS is 0

speed from GNSS is 0.0

...

//these attempts go ahead until the response of the AT#LWM2MOBJGET=0,6,0 will be different from zero

...

waiting for fix

sending: AT#LWM2MR=0,3,0,13,0

response:#LWM2MR: 1637234664

sending: AT#LWM2MOBJGET=0,6,0
```

```
response:{"n":"/6/0/","e":[{"n":"0","v":39.227080}, {"n":"1","v":9.069071}, {"n":"2","v":-20.177584}, {"n":"3","v":3.535534}, {"n":"4","sv":"AAAAAA=="}, {"n":"5","v":1637234668}, {"n":"6","v":0.000000}]}
```

latitude from GNSS is 39.22708

longitude from GNSS is 9.069071

altitude from GNSS is -20.177584

radius from GNSS is 3.535534

velocity from GNSS is AAAAAA==

timestamp from GNSS is 1637234668

speed from GNSS is 0.0

fix found

DATA sending to OneEdge portal

sending: AT#LWM2MENA=1

OK

LWM2M-TLT:"REGISTERING",SSID=99,"coap://api-dev.devicewise.com"

LWM2M-TLT:"REGISTERED",SSID=99,"coap://api-dev.devicewise.com"

Wait for 20 seconds to portal data read

sending: AT#LWM2MENA=0

OK

LWM2M-TLT:"DEREGISTERING",SSID=99,"coap://api-dev.devicewise.com"

LWM2M-TLT:"DEREGISTERED",SSID=99,"coap://api-dev.devicewise.com"

5.3.4.4.2. MEx10G1 Environment

Since the procedure to manage the GNSS data on MEx10G1 devices is a bit different than ME910C1, the Python script has been modified according to paragraph [5.3.4.2.3](#).

The report below shows the output of the script in this case.

OPEN THE SERIAL PORT

USE DEVICE PROFILE WITH READ ON REGISTER ON OBJ 6/0

sending: AT\$GPSCFG=3,0

OK

START LOOP for data upload

LOOP 1

sending: AT#LWM2MW=0,33211,0,0,0,1

OK

GNSS Session is ongoing

START LOOP for fix

waiting for fix

sending: AT#LWM2MR=0,3,0,13,0

response:#LWM2MR: 1637234650

sending: AT#LWM2MOBJGET=0,6,0

response:{"n":"/6/0/","e":[{"n":"0","v":0.000000}, {"n":"1","v":0.000000}, {"n":"2","v":0.000000}, {"n":"3","v":0.000000}, {"n":"4","sv":"AAAAAA=="}, {"n":"5","v":0}, {"n":"6","v":0.000000}]}

latitude from GNSS is 0.0

longitude from GNSS is 0.0

altitude from GNSS is 0.0

radius from GNSS is 0.0

velocity from GNSS is AAAAAA==

timestamp from GNSS is 0

speed from GNSS is 0.0

...

//these attempts go ahead until the response of the AT#LWM2MOBJGET=0,6,0 will be different from zero

...

waiting for fix

sending: AT#LWM2MR=0,3,0,13,0

response:#LWM2MR: 1637234664

sending: AT#LWM2MOBJGET=0,6,0

response:{"n":"/6/0/","e":[{"n":"0","v":39.227080}, {"n":"1","v":9.069071}, {"n":"2","v":-20.177584}, {"n":"3","v":3.535534}, {"n":"4","sv":"AAAAAA=="}, {"n":"5","v":1637234668}, {"n":"6","v":0.000000}]}

latitude from GNSS is 39.22708

longitude from GNSS is 9.069071

altitude from GNSS is -20.177584

radius from GNSS is 3.535534

velocity from GNSS is AAAAAA==

timestamp from GNSS is 1637234668

speed from GNSS is 0.0

fix found

sending: AT#LWM2MW=0,33211,0,0,0,0

OK

WWAN Session is ongoing

sending: AT#LWM2MENA=1

OK

sending: AT#LWM2MENA=0

OKLWM2M-TLT:"DEREGISTERING",SSID=99,"coap://api-dev.devicewise.com"LWM2M-TLT:"DEREGISTERED",SSID=99,"coap://api-dev.devicewise.com"

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Do not disassemble the product; any mark of tampering will compromise the warranty validity. We recommend following the instructions of the hardware user guides for correct wiring of the product. The product has to be supplied with a stabilized voltage source and the wiring has to be conformed to the security and fire prevention regulations. The product has to be handled with care, avoiding any contact with the pins because electrostatic discharges may damage the product itself. Same cautions have to be taken for the SIM, checking carefully the instruction for its use. Do not insert or remove the SIM when the product is in power saving mode.

The system integrator is responsible for the functioning of the final product. Therefore, the external components of the module, as well as any project or installation issue, have to be handled with care. Any interference may cause the risk of disturbing the GSM network or external devices or having an impact on the security system. Should there be any doubt, please refer to the technical documentation and the regulations in force. Every module has to be equipped with a proper antenna with specific characteristics. The antenna has to be installed carefully in order to avoid any interference with other electronic devices and has to guarantee a minimum distance from the body (20 cm). In case this requirement cannot be satisfied, the system integrator has to assess the final product against the SAR regulation.

The equipment is intended to be installed in a restricted area location.

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The European Community provides some Directives for the electronic equipment introduced on the market. All of the relevant information is available on the European Community website:

https://ec.europa.eu/growth/sectors/electrical-engineering_en

7. GLOSSARY

ADC	Analog – Digital Converter
CLK	Clock
CMOS	Complementary Metal – Oxide Semiconductor
CS	Chip Select
DAC	Digital – Analog Converter
DTE	Data Terminal Equipment
ESR	Equivalent Series Resistance
GNSS	Global Navigation Satellite System
HS	High Speed
HSDPA	High Speed Downlink Packet Access
HSIC	High Speed Inter Chip
HSUPA	High Speed Uplink Packet Access
I/O	Input Output
MISO	Master Input – Slave Output
MOSI	Master Output – Slave Input
MRDY	Master Ready
PCB	Printed Circuit Board
RTC	Real Time Clock
SIM	Subscriber Identification Module
SPI	Serial Peripheral Interface
SRDY	Slave Ready
TTSC	Telit Technical Support Centre
UART	Universal Asynchronous Receiver Transmitter
UMTS	Universal Mobile Telecommunication System
USB	Universal Serial Bus
VNA	Vector Network Analyzer
VSWR	Voltage Standing Wave Ratio
WWAN	Wireless Wide Area Network

8. DOCUMENT HISTORY

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